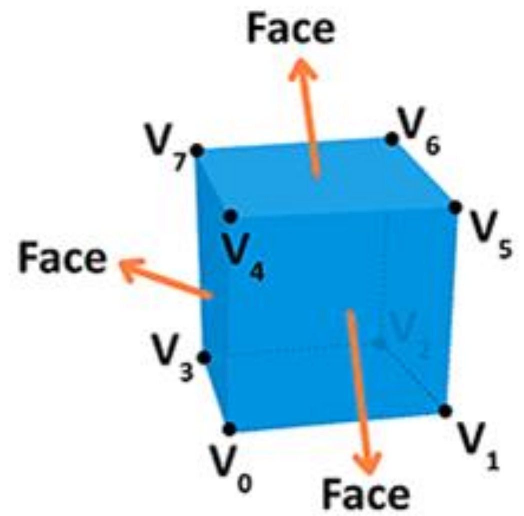
Three Dimensional Basics (3D)

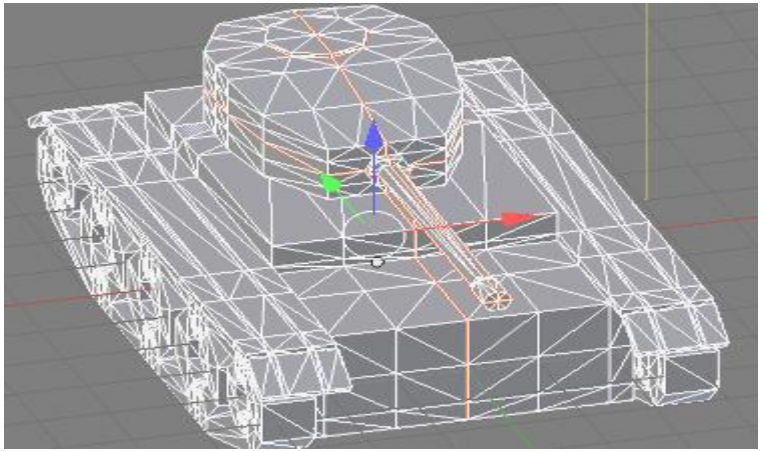
Objects

- A **Vertex** is a point in space having its own 3D position in the coordinate system
- **Position**: Identifies it in a 3D space (x, y, z).
- **Color**: Holds an RGBA value (R, G and B for the red, green, and blue channels, alpha for transparency — all values range from 0.0 to 1.0).
- **Normal:** A way to describe the direction the vertex is facing.
- **Texture**: A 2D image that the vertex can use to decorate the surface it is part of instead of a simple color.



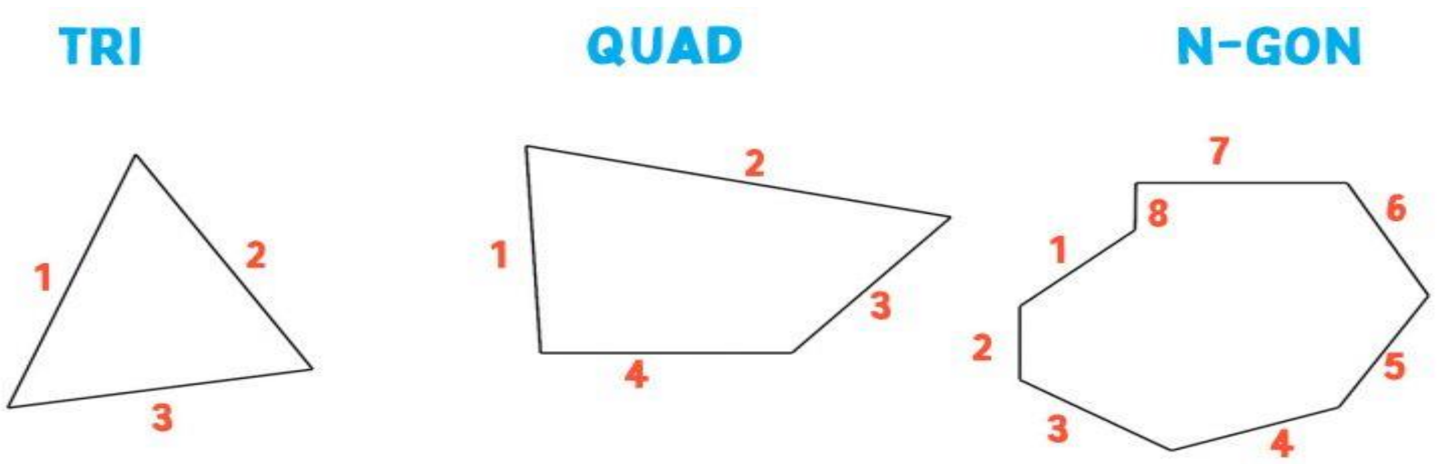
Mesh

- A mesh is a collection of vertices, edges, and faces that define the shape of a 3D object. The most popular type of polygon mesh used in computer graphics is a Triangle Mesh. Any object, either a mountain or a game character, can be modelled with Triangle meshes.



Faces: Triangle, Quad, N-Gon

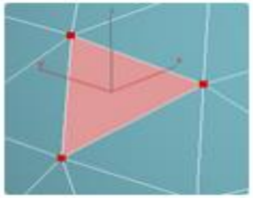
- A face can have a minimum of 3 vertices. This is then a **triangle**. If you have four vertices it is called a **quad**, and if you have more than four vertices, then it is simply a **general polygon**.
- When you have 5 or more edges around a face, it’s referred to as an **N-gon**.



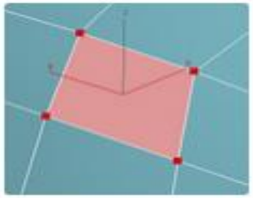
Tris, Quads & N-Gons

- A triangular polygon is referred to as a ***tri*** or ***triangle***, and is a simple three-sided polygon. It has exactly 3 vertices at its corners and 3 edges connecting those points. This is the smallest configuration required to make a polygonal face.
- A square or rectangular polygon is referred to as a ***quad*** *or* ***quadrilateral*** polygon, and is a four-sided polygon. It has exactly 4 vertices at the corners connected by 4 edges. This is the most desired type of polygon when creating digital models, and many artists like to build their objects using nothing but quads to help make their work more appealing to customers in complex pipelines.
- An ***N-Gon*** is a polygon with more than four vertices and edges. This kind of polygon is not desired and should be avoided. Due to its geometric properties, an N-Gon can **always** be divided into quads, tris, or a combination of the two; so they are always easy to remove by adding connecting edges between the border vertices.

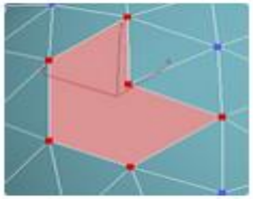
https://resources.turbosquid.com/training/modeling/tris-quads-n-gons/



Triangular Poly (3 Sides)

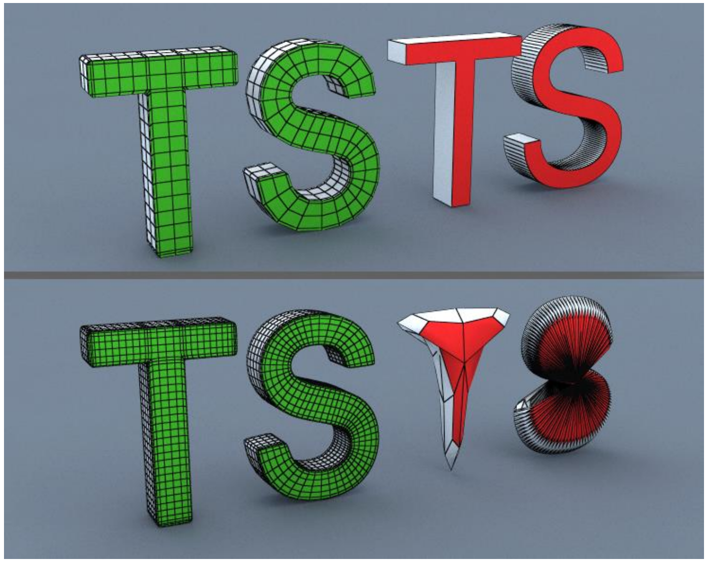


Quad Polygon (4 Sides)

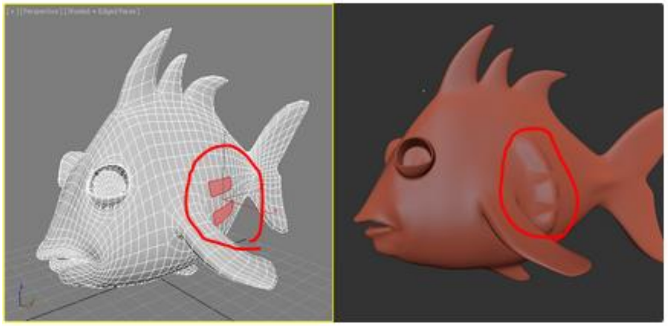


N-Gon Poly (More than 4 Sides)

Why N-Gons Should Be Avoided



Ngons (marked in Red) don’t subdivide cleanly or predictably



NGons in 3ds Max — Unable to Sculpt cleanly in Mudbox

Difficulty when subdividing

- N-Gons subdivide تقسيم الشكل لأشكال أصغر poorly and produce unwanted topology شكل هندسي, even on flat surfaces. While it may be quicker to model using N-Gons, it is a bad habit to get into as they disrupt the edge flow of your model, making selection of edge loops more difficult.
- Having N-Gons often leads to strange rendering or smoothing artifacts that are hard or impossible to eliminate.
- Instead of leaving N-Gons around in your model, it is better to avoid them altogether since quads and tris will always subdivide properly.

Unwanted behavior when importing and exporting

- Some applications do not handle N-Gons well or at all.
- Popular sculpting النحت products like ZBrush, Mudbox and 3D Coat will bring up alert messages when an artist tries to import a model with N-Gons present, and sculpting across those surfaces will produce unwanted results.
- For customers, Ngons can cause major issues when trying to use your model in multiple environments and as such, need to be avoided.

Non-Planar possibility

- The triangle can only be on one plane. Existing on only one plane is called being planar. Quads and n-gons can be, and ideally are, only planar. But you can bend up a corner and they can become non-planar. Now, not all applications can work with non-planar polygons. If a polygon is non-planar, it can and will sometimes automatically, be broken down into the simpler and more digestible triangles.

Planar and Non-Planar Faces

In planar faces all points lay on the same plane

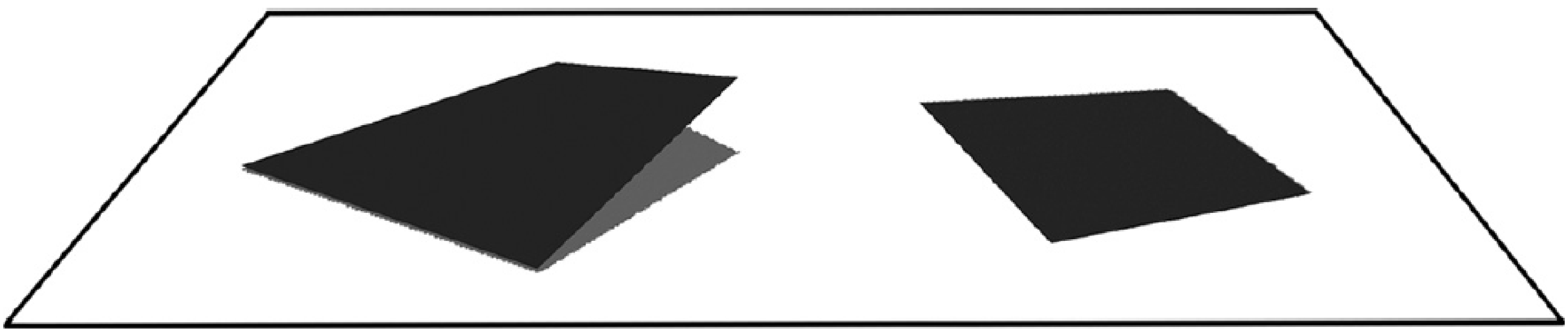
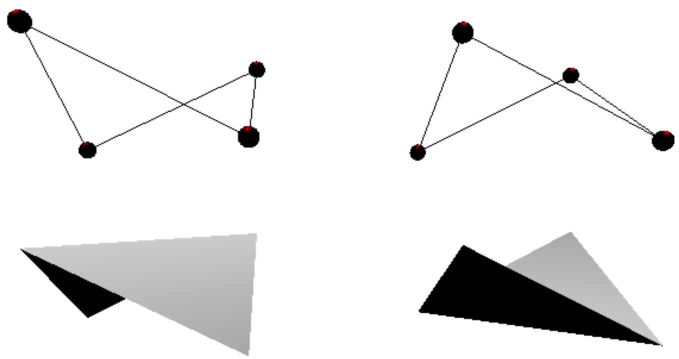
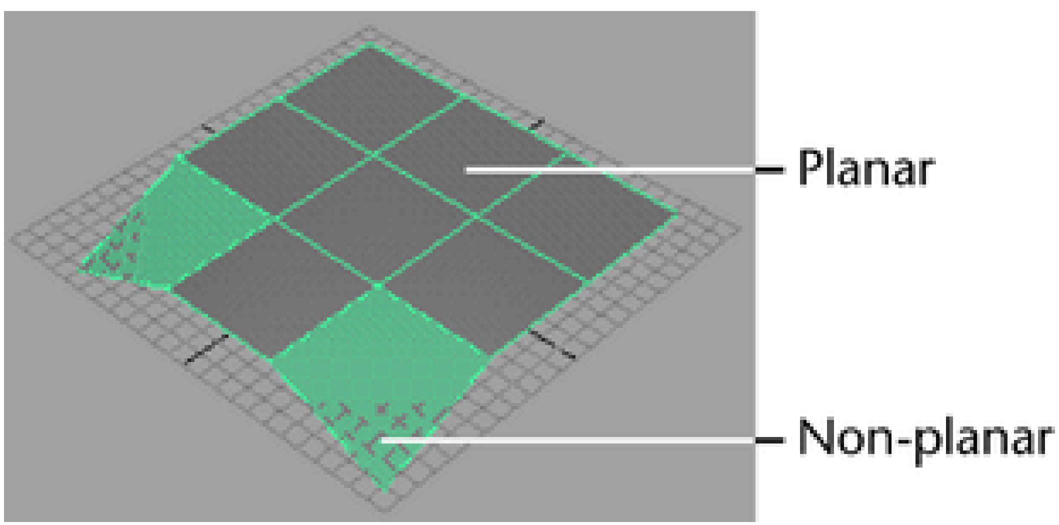
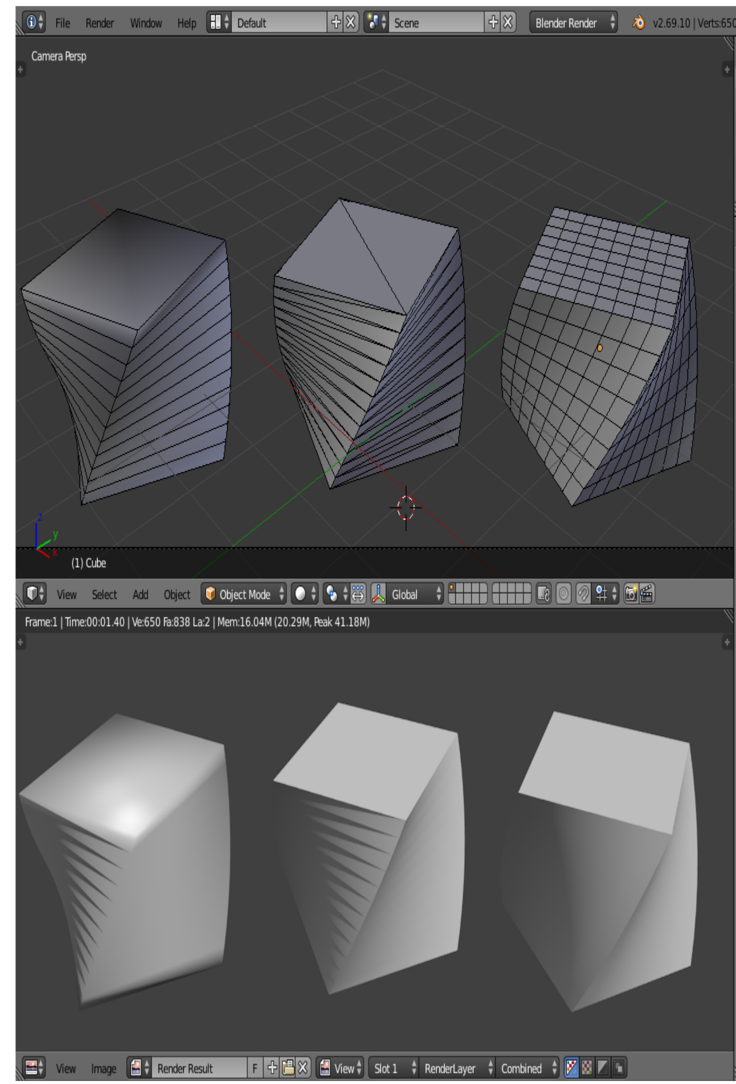


Figure 3.3 Non-planar (left) and planar (right) polygons.

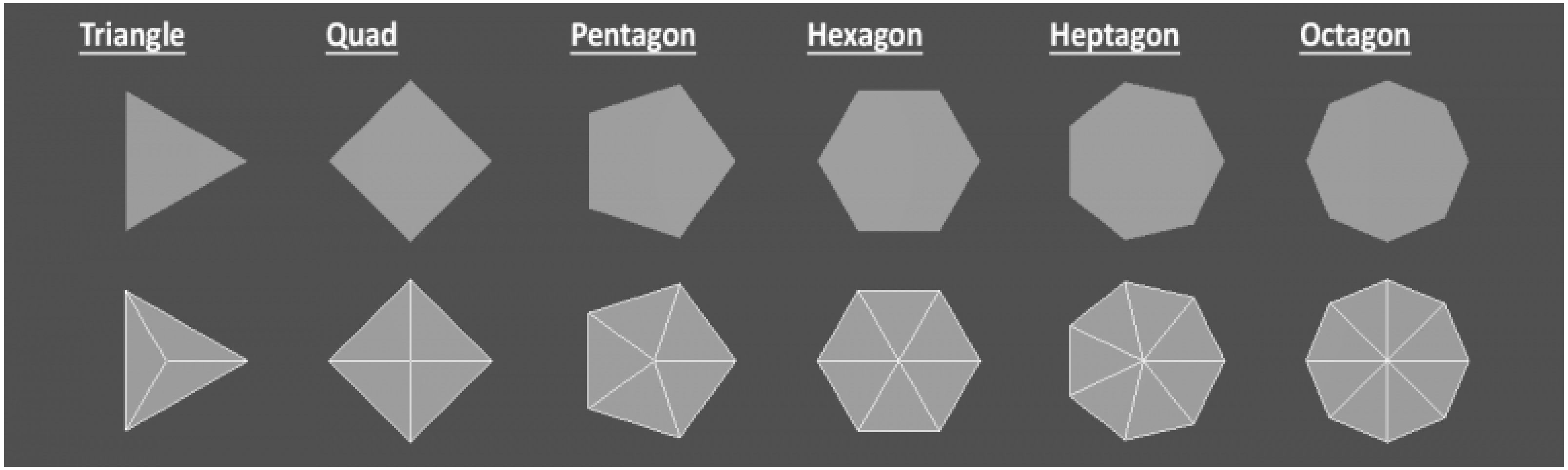
Examples of Non-Planar faces cases







Examples for Splitting N-Gon

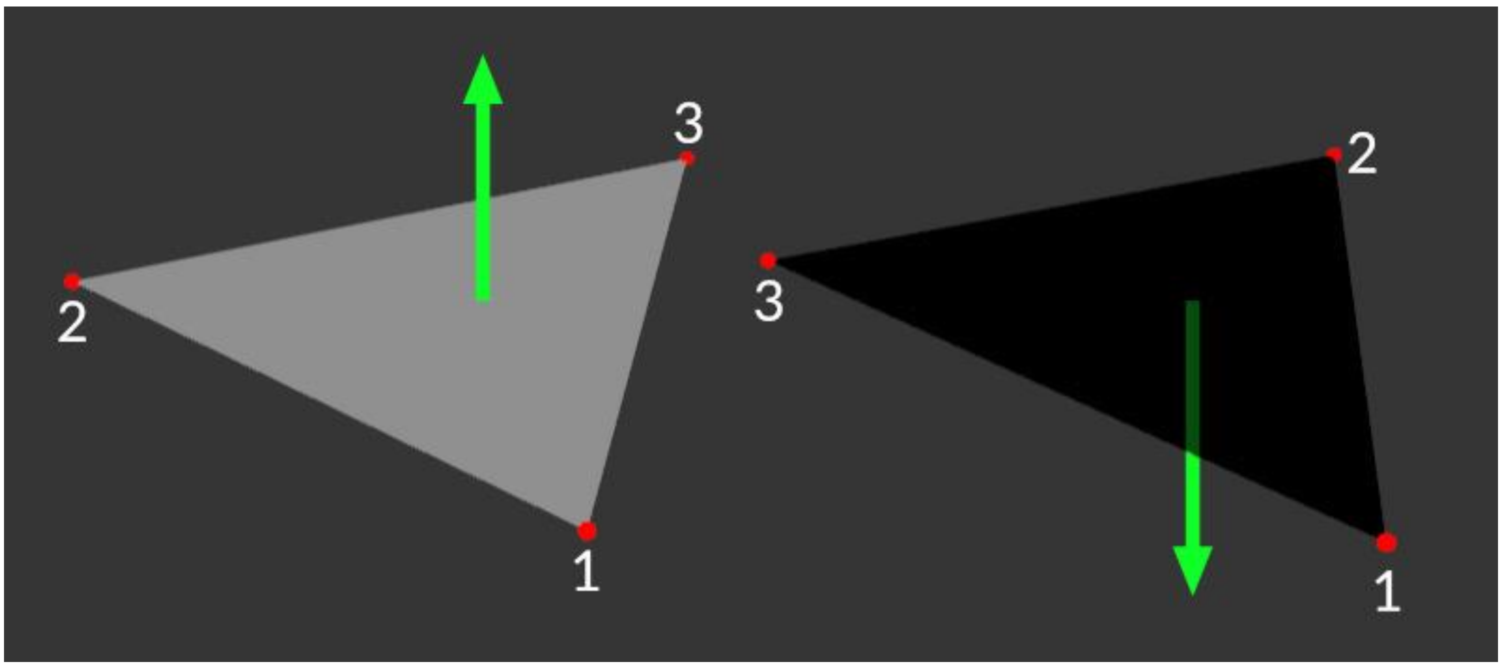


Normals

- There are two types of normal:
  - Face normals :
    - single normal for entire face
  - vertex normal:
    - A normal for each vertex
  - Interpolated normal:
    - A calculated normal based on values of other vertex normal

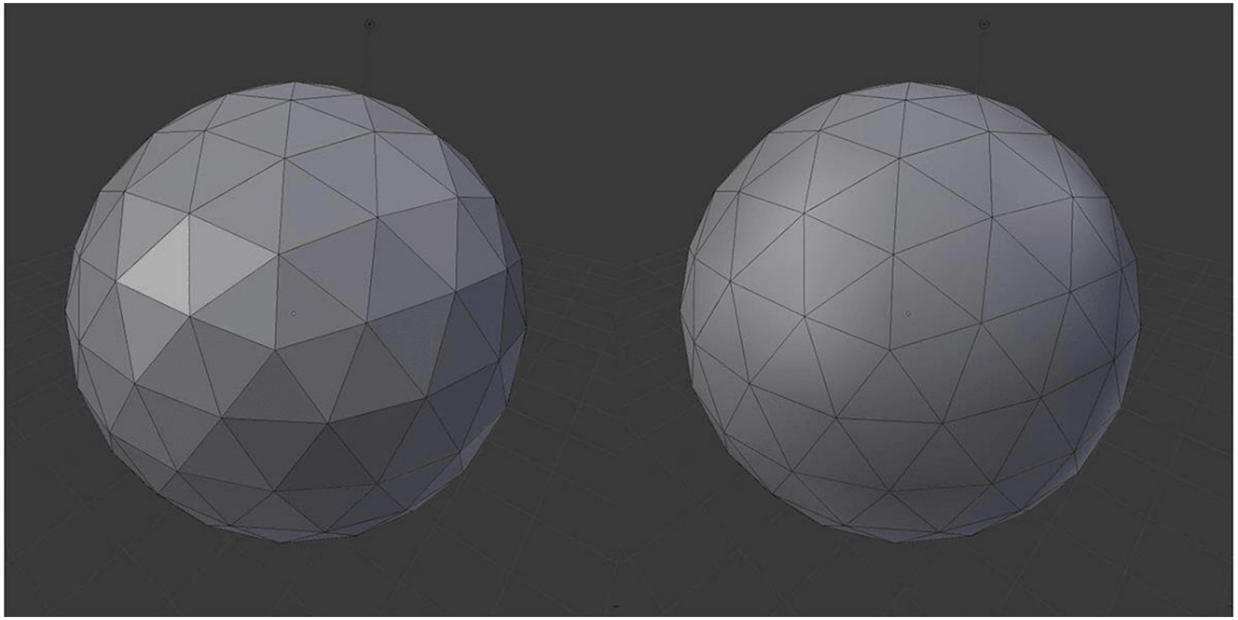
Face normals

- A normal is just a **hidden vector**, basically a direction without a specified origin.

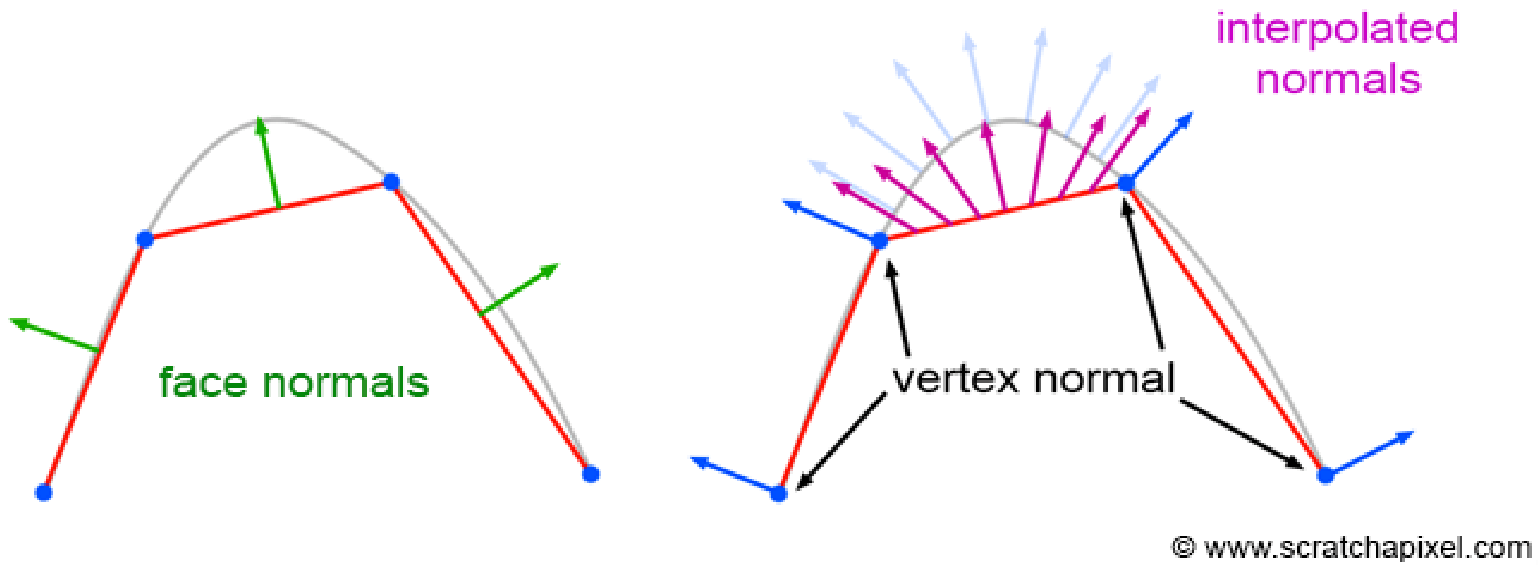


Vertex Normal

- Those 2 models consist of the same vertex positions,
- yet look totally different when rendered. How is that possible?
- we can also give it a hint on how the surface is sloping يميل in that exact position.
- The hint is in the form of the normal of the surface at that specific point on the model.



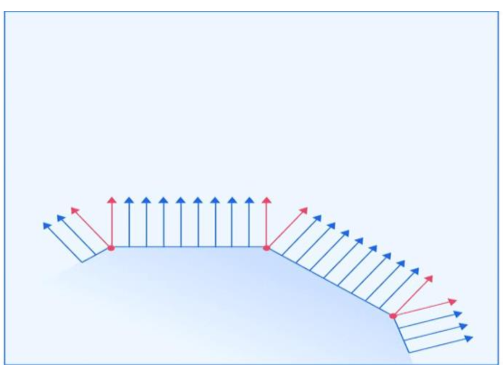
Interpolated (Averaged) Normals

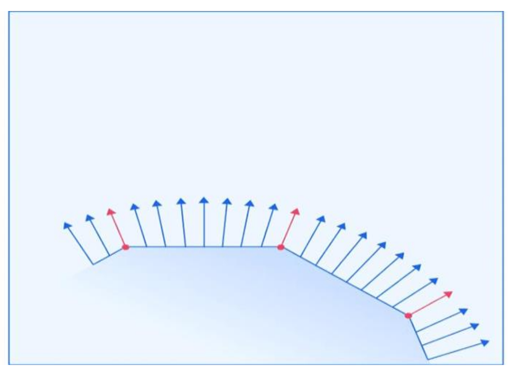


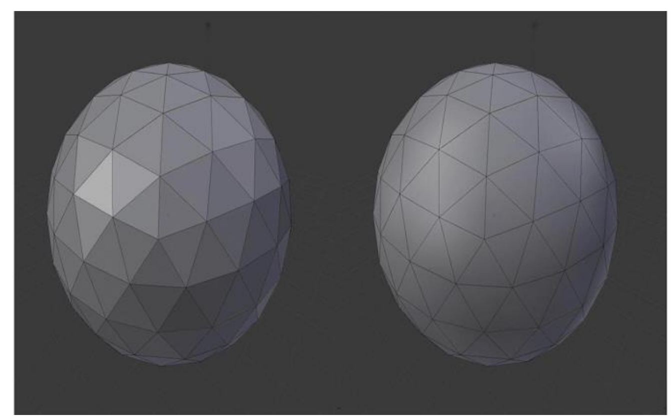
- The technique is illustrated in the image above. Vertex normals are defined at the **faces vertices**.
- Interpolated normal are calculated based on the values of other vertex normal
- Can be used by the renderer to change light angle

Vertex Normal-2

- The left and right surface correspond to the left and right ball in the previous image, respectively.
- The red arrows represent vertex normals that are
- blue arrows represent the renderer’s calculations of how the normal should look for all the points between the vertices.



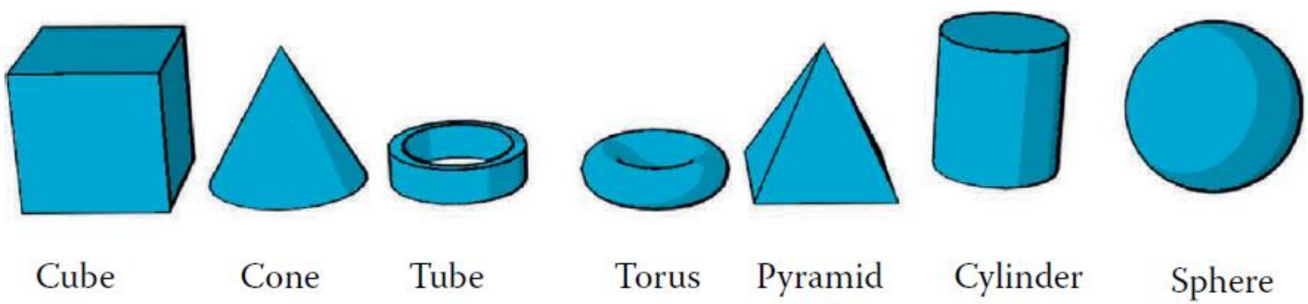




Basic 3D Shapes

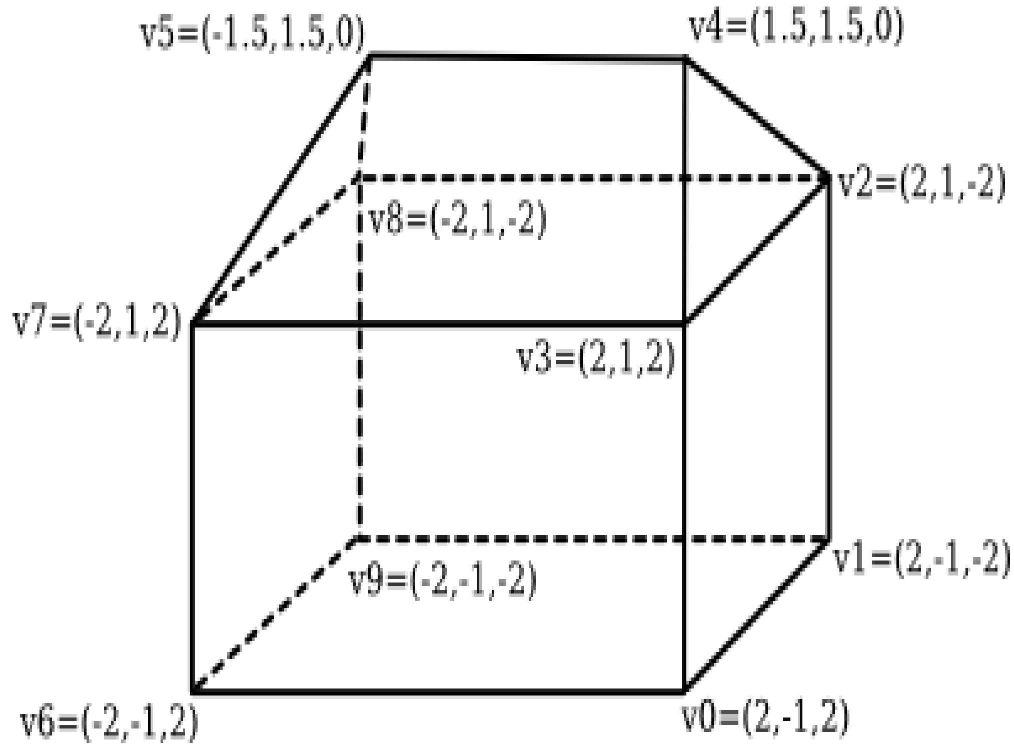
Cube or box

- Cone
- Tube
- Torus
- Pyramid
- Cylinder
- Sphere



Indexed Face Sets

- The vertex list for this polyhedron has the form
- Vertex #0. (2, -1, 2)
- Vertex #1. (2, -1, -2)
- Vertex #2. (2, 1, -2)
- Vertex #3. (2, 1, 2)
- Vertex #4. (1.5, 1.5, 0)
- Vertex #5. (-1.5, 1.5, 0)
- Vertex #6. (-2, -1, 2)
- Vertex #7. (-2, 1, 2)
- Vertex #8. (-2, 1, -2)
- Vertex #9. (-2, -1, -2)



Indexed Face Sets-p2

To describe one of the polygonal **faces of a mesh**, we just have to list its vertices

Face #0: (0, 1, 2, 3)

Face #1: (3, 2, 4)

Face #2: (7, 3, 4, 5)

Face #3: (2, 8, 5, 4)

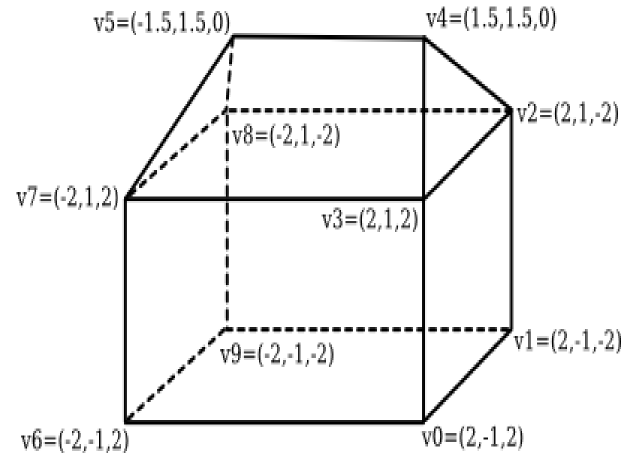
Face #4: (5, 8, 7)

Face #5: (0, 3, 7, 6)

Face #6: (0, 6, 9, 1)

Face #7: (2, 1, 9, 8)

Face #8: (6, 7, 8, 9)



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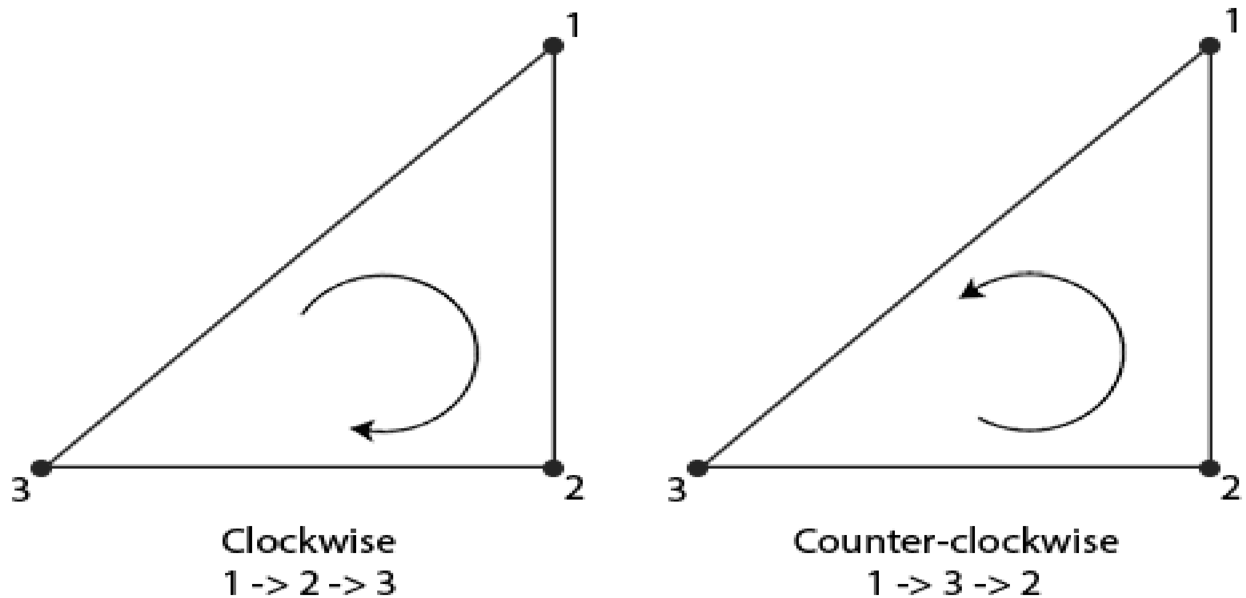
- Vertex #5. (-1.5, 1.5, 0)
- Vertex #6. (-2, -1, 2)
- Vertex #7. (-2, 1, 2)
- Vertex #8. (-2, 1, -2)
- Vertex #9. (-2, -1, -2)

Winding order



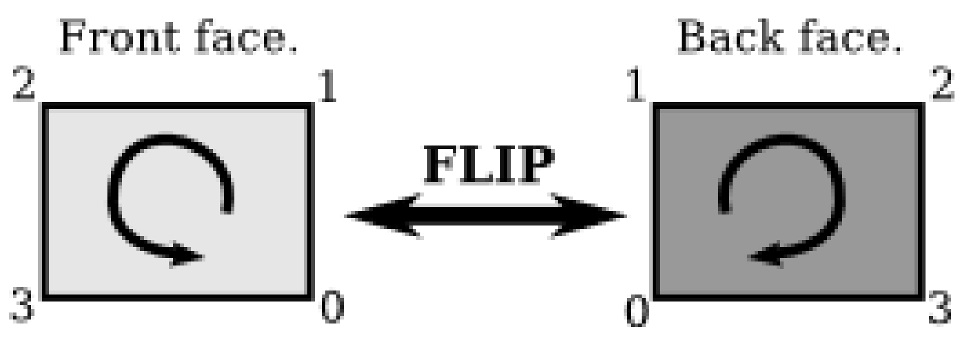
- When we define a set of triangle vertices we're defining them in a certain winding order that is either clockwise or counter-clockwise. Each triangle consists of 3 vertices and we specify those 3 vertices in a winding order as seen from the center of the triangle.

Winding order-2

- Some software use (CW)= (Clockwise winding), Other use (CCW)= (Counter Clockwise winding) :
    - Unity uses CW.
    - Blender defaults to CW,
    - Maya and 3D Studio both currently default to CCW
- Now you define a face following a certain winding order - say CCW - to be the front face, the other side to be the back face. Which winding order corresponds to the front is completely arbitrary and depends on the game engine you use. However for some reason this fact is seldom documented and you have to find out yourself through testing.

Back face

- It turns out that it is often convenient to consider one of those faces to be the "front face" of the polygon and one to be the "back face."
- The usual rule is that the vertices of a polygon should be listed in counter-clockwise order when looking at the front face of the polygon.



Vertex order 0,1,2,3 is counter-clockwise from the front and clockwise from the back

3D file formats

- The basic purpose of a 3D file format is to store information about 3D models as plain text or binary data. In particular, they encode the 3D model's *geometry*, *appearance*, *scene*, and *animations*. There are a lot of 3d formats like (fbx, 3ds, blend, obj, …).
- **FBX** is one of the most used 3d formats.

Open source 3D modeling Tools (ref)

Software	Platform	Specialization	Use Cases
Tinkercad	Browser	Direct Modeling, OpenSCAD	Education
Vectary	Browser	Mesh Modeling, Parametric Modeling	Product Design, Graphic Design, Animation
Meshmixer	Windows, macOS	Direct Modeling, Sculpting, Optimization	3D Printing Optimization/ Repair
SculptGL	Browser, Windows, ios, android	Sculpting	Digital Art, Animation, Game Design
ZBrushCoreMini	Windows, macOS	Sculpting	Digital Art, Animation, Game Design
SketchUp Free	Browser	Direct Modeling	Architecture, Product Design
Wings 3D	Windows, MacOS, Linux	Mesh Modeling	Architecture, Product Design
Leopoly	Browser	Sculpting	Digital Art, Animation, Game Design
BlocksCAD	Browser	OpenSCAD	Education
Blender	Windows, MacOS, Linux	Sculpting, Mesh Modeling	Digital Art, Animation, Game Design

Open source 3D modeling Tools للاطلاع فقط

The Best 8 Free and Open Source 3D Modeling Software						
3D Modeling Software	**License**	**Initial Release**	**3D Rendering Support**	**User Level**	**Platform**	**File Formats**
Blender	GPLv2+	1998	Yes	Professional	Windows, Mac,Linux	3ds, dae, fbx,dxf,obj,x,lwo,svg,ply,STL,vrml97,x3d
Sculptris	Freeware	2009	No	Intermediate	Windows, Mac	obj, goz
Wings 3D	BSD	2018	Yes	Beginner	Windows, Mac, Linux	3ds,fbx,obj,dae, lwo,wrl,rwx,STL,x,xml
Seamless 3D	MIT	2001	No	Beginner	Windows	Import/Export files like VRML, X3D Obj
Dust3D	MIT		Yes	Game developers	Cross platform-Windows, Linux, MacOS	FBX,glTF
Mandelbulb 3D	Freeware	2009	Yes	Beginner	Cross-Platform	M3I image + Parameter, .M3P Parameter
SketchUp	Freemium	2000	Yes	Advanced	Windows, Mac	skp,stl,png
Onshape	Commercial Software	2015	Yes	Professional	Windows, Mac,Linux,Chromebook, iOS &Android	Parasolid,ACIS,STEP, IGES,SolidWorks, Inventor,Rhino, OBJ,NX,DXF,DWG, PDF, STL

Blender

Blender is one of the best free open source 3D modeling software that can be beneficial for:

1. 3D modeling,
2. rendering,
3. game creation,
4. video editing,
5. compositing,
6. rigging,
7. animation,
8. motion tracking and much more.

The active community of users that consist of studios and hobbyists support the software.

3D Modelling Techniques

- 3D models can be created in many different ways. The choice of modelling technique depends on the requirements and the complexity of the object. The following list describes some of popular 3D modelling techniques

-Modeling Workflows

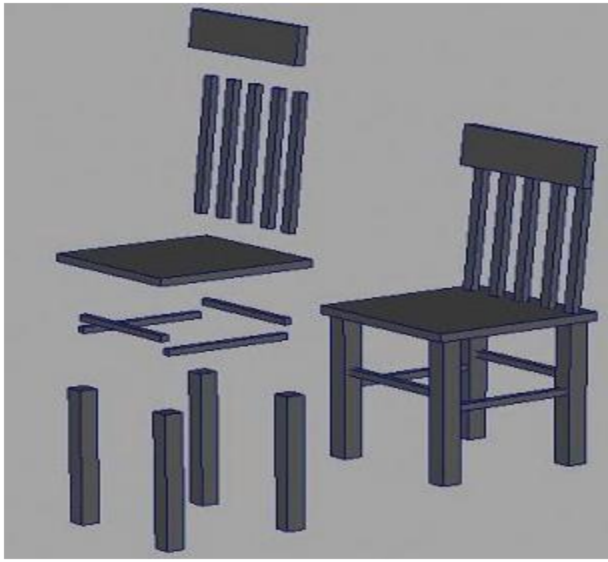
1. **From-Scratch modeling** :
   In from-scratch modeling, you lay out each vertex and draw each polygon one by one until the entire model is completed

Modeling Workflows (2)

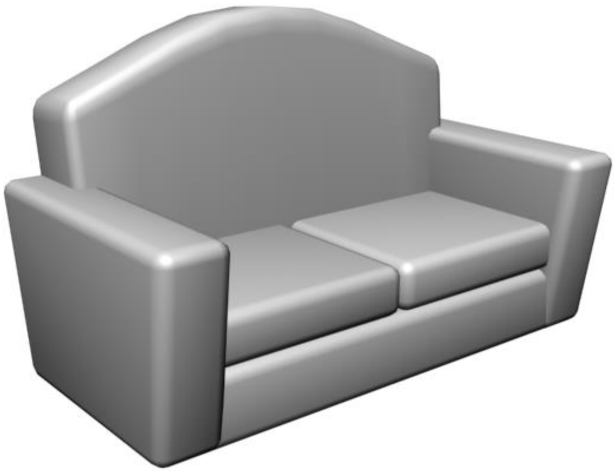
2- Primitive modeling :

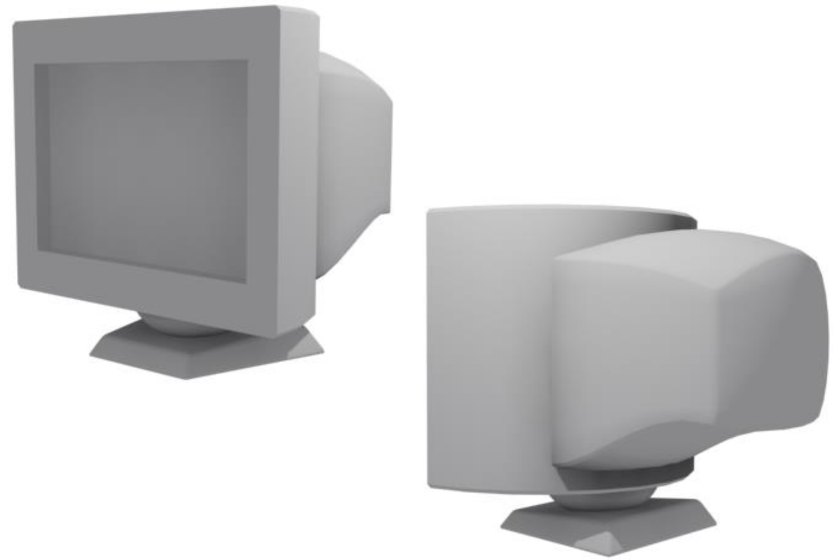
start with basic shapes (cylinder, torus حلقة, sphere, cube,…) and add changes till it becomes your target shape

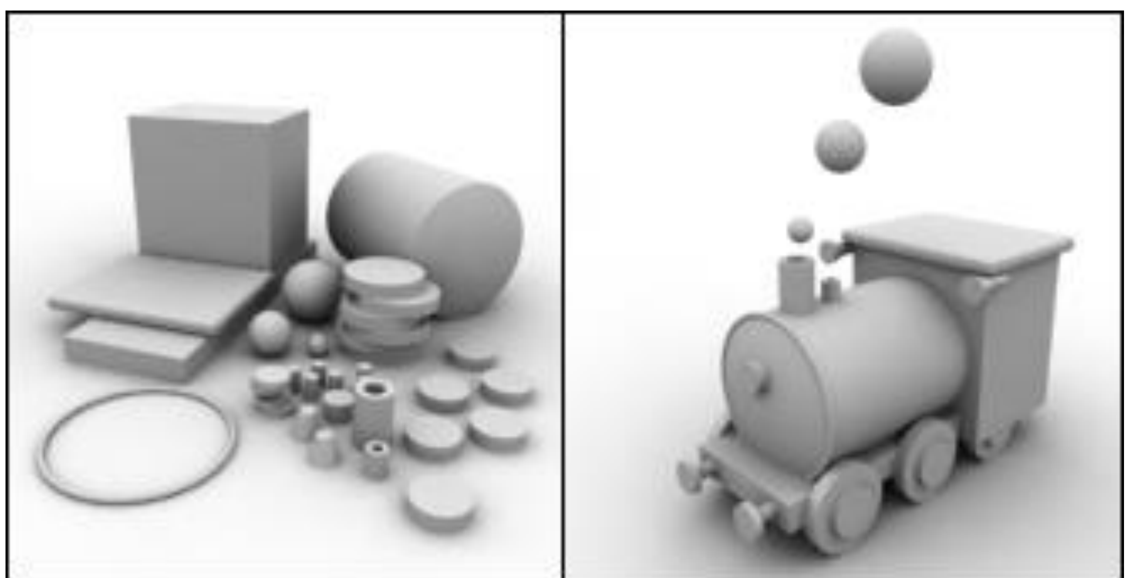
(great for hard-surface models such as tables, chairs, pencils, picture frames)



More Examples on Primitive Modeling

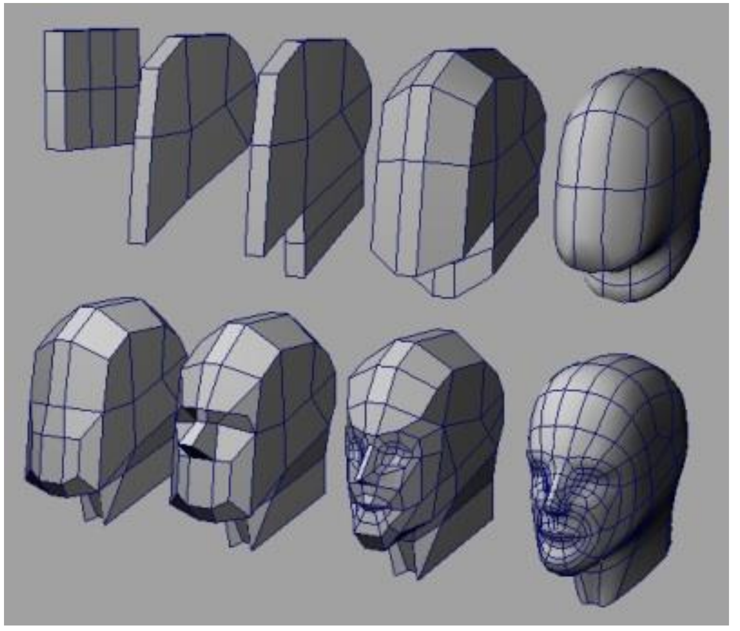




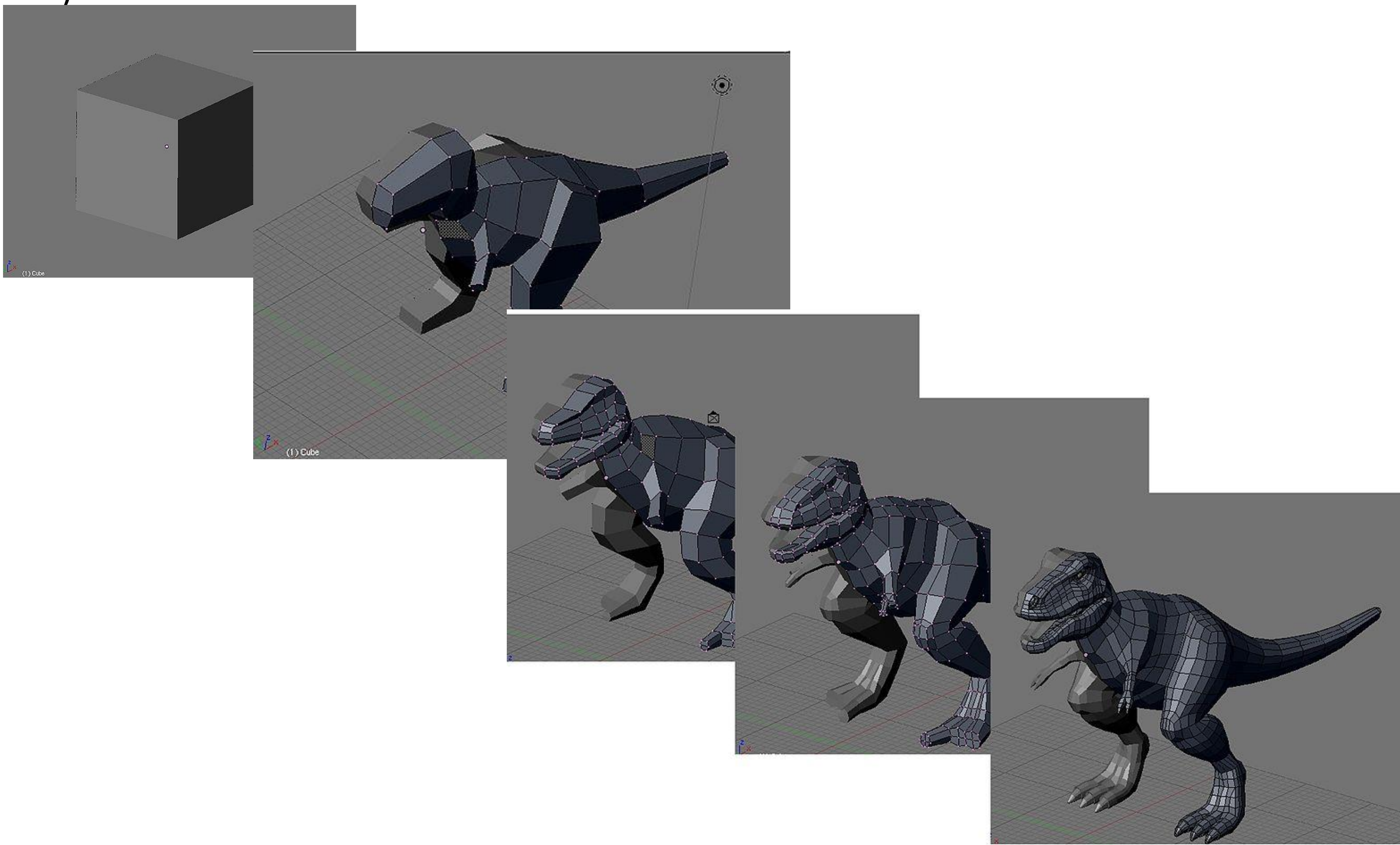


Modeling Workflows (3)

- 3-Box Modeling:
  - Box modeling is a popular approach to character modeling
  - quick creation of basic shapes.
  - start with a **cube** and extrude arms, fingers, legs, toes, and a head.
  - You then add detail to the whole shape by splitting and refining the model

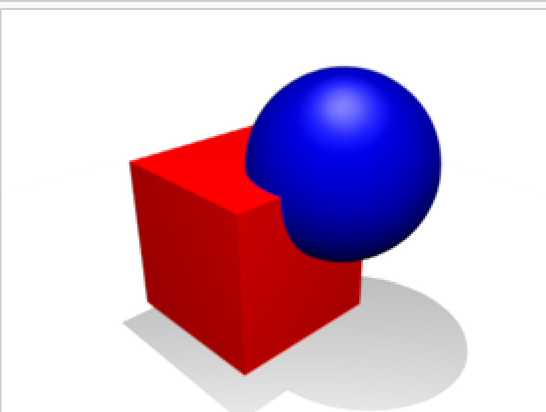


Example on Box Modeling
(https://en.wikibooks.org/wiki/Blender_3D:_Noob_to_Pro/Box_Modeling
)

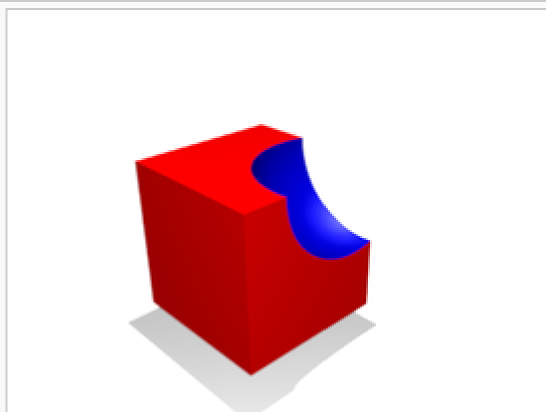


Modeling Workflows (4)

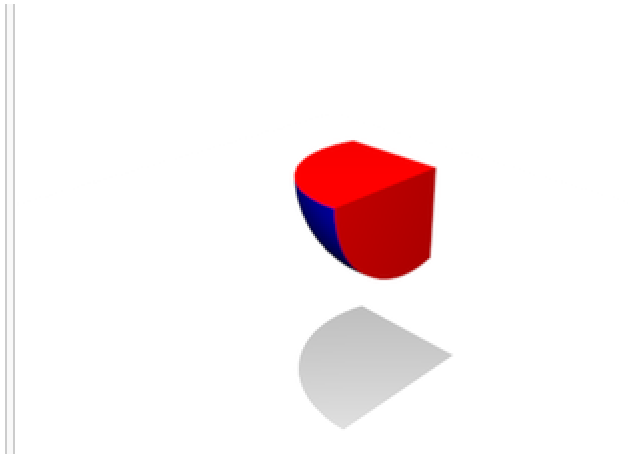
- **4-Boolean modeling (Constructive Solid Geometry):**
  - additive or subtractive method of changing the geometry of one object by introducing another object and running a Boolean function:
    - Union
    - Intersection
    - Difference



Union
Merger of two objects into one

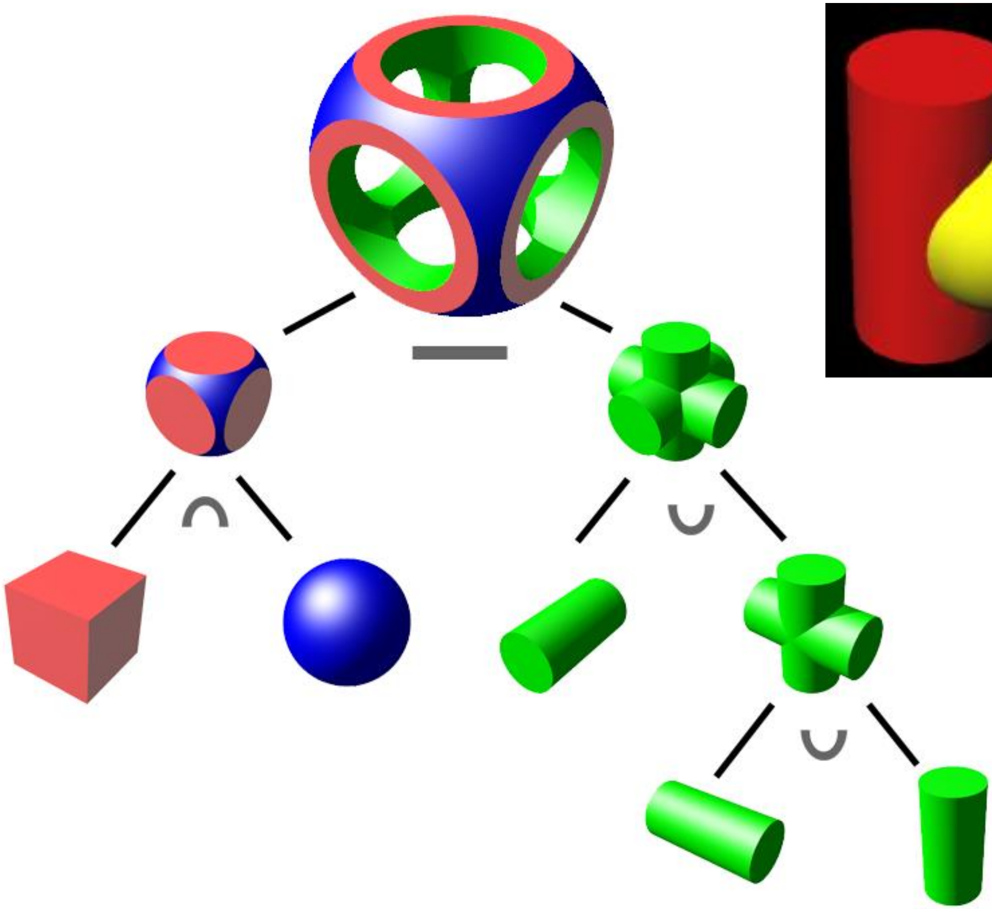


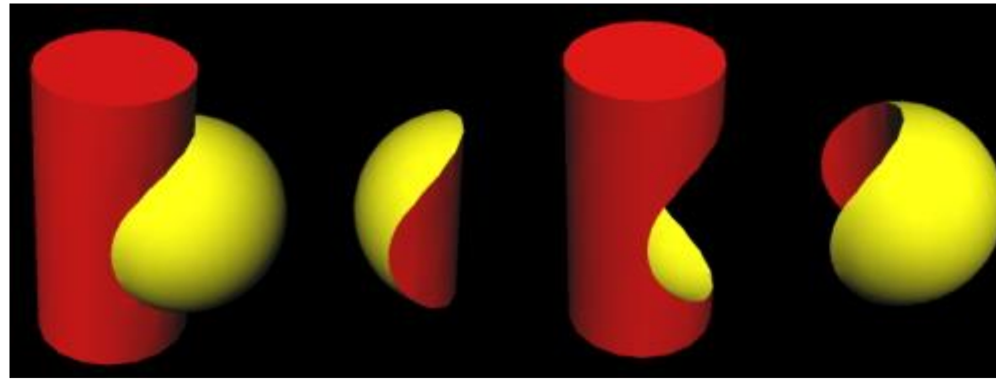
Difference
Subtraction of one object from another

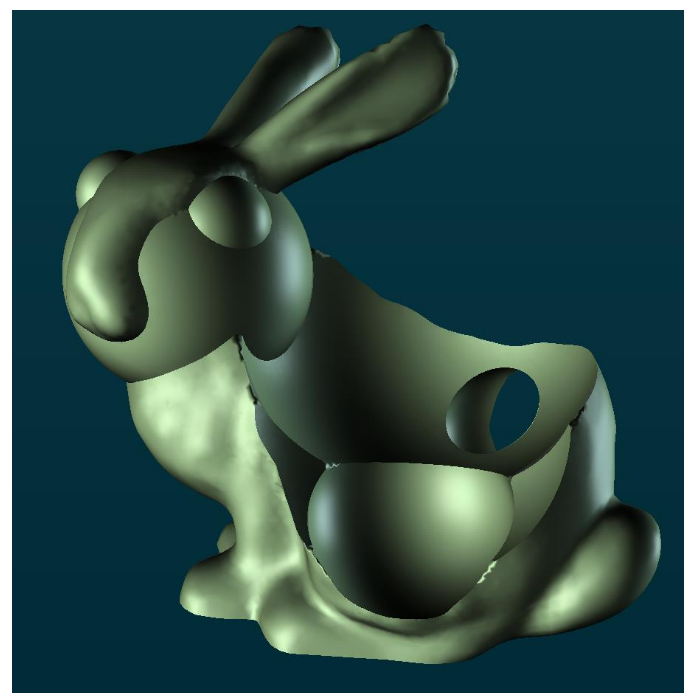


Intersection
Portion common to both objects

Complex shapes using Boolean Modeling





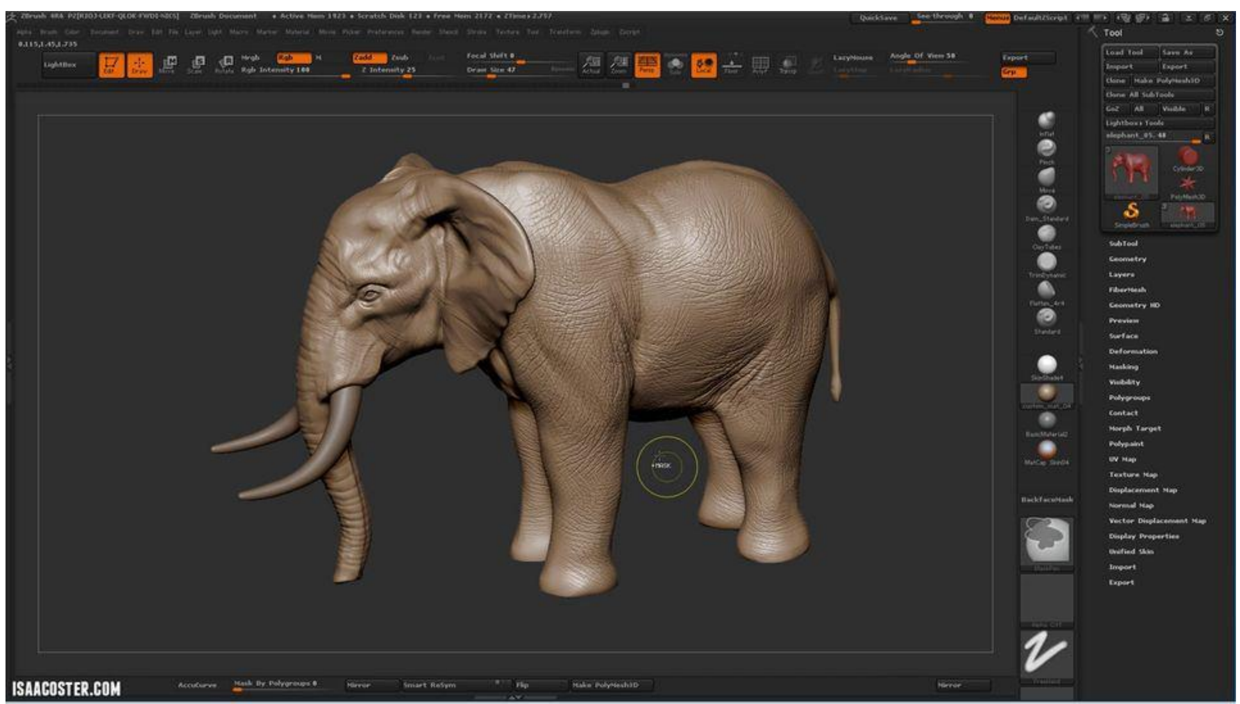


Modeling Workflows (5)

- **5-Laser Scanning**
  - starts with a real object that you digitally laser-scan to create the geometric surfaces.
  - The scanning process can be a quick and easy task,
  - but the geometry created is always very Complex

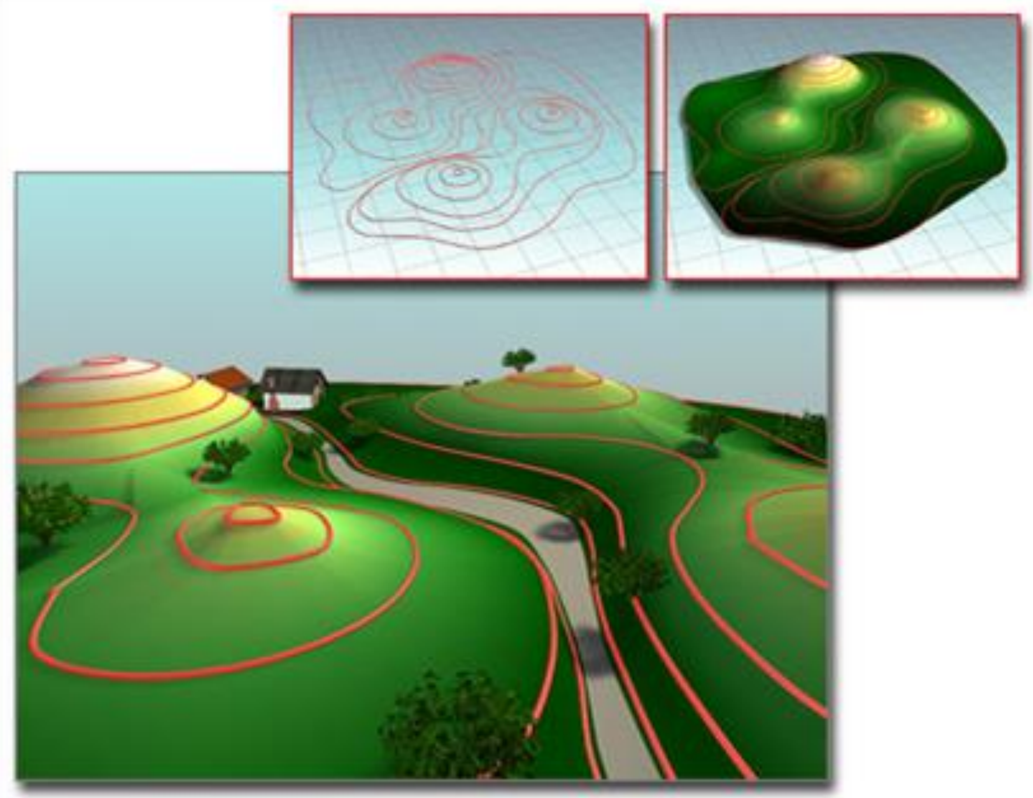
Modeling Workflows (6)

- **6- Digital Sculpting:**
  - artists to use a **massive** polygon resolution to create a surface that can be **pushed** and **pulled** like clay صلصال



7-Terrain التضاريس

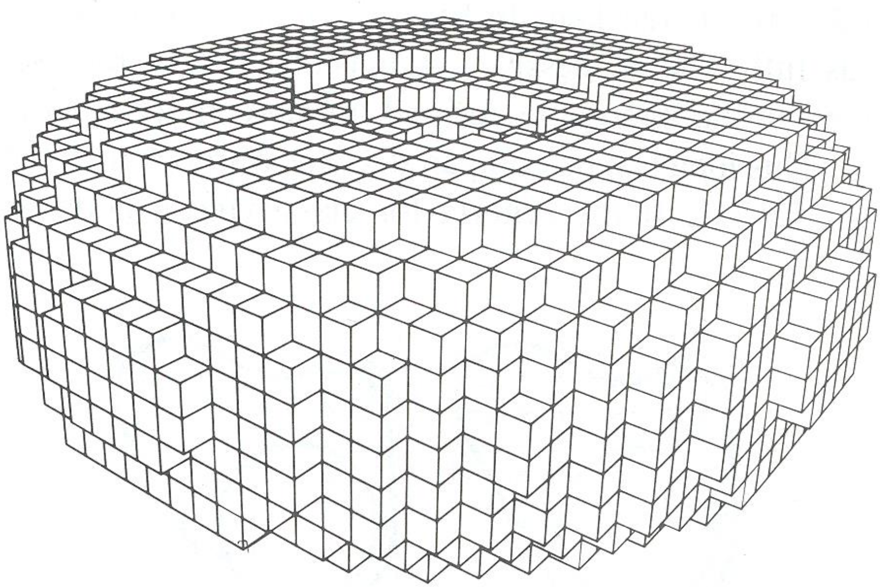
- Some 3D applications generate these objects from contour line data.
- select **editable curves** representing elevation الارتفاعات and create a mesh surface over the contours.



Using contours to build a terrain
Upper left: The contours
Upper right: The terrain object
Lower left: Terrain object used as the basis of a landscape

8-Voxels

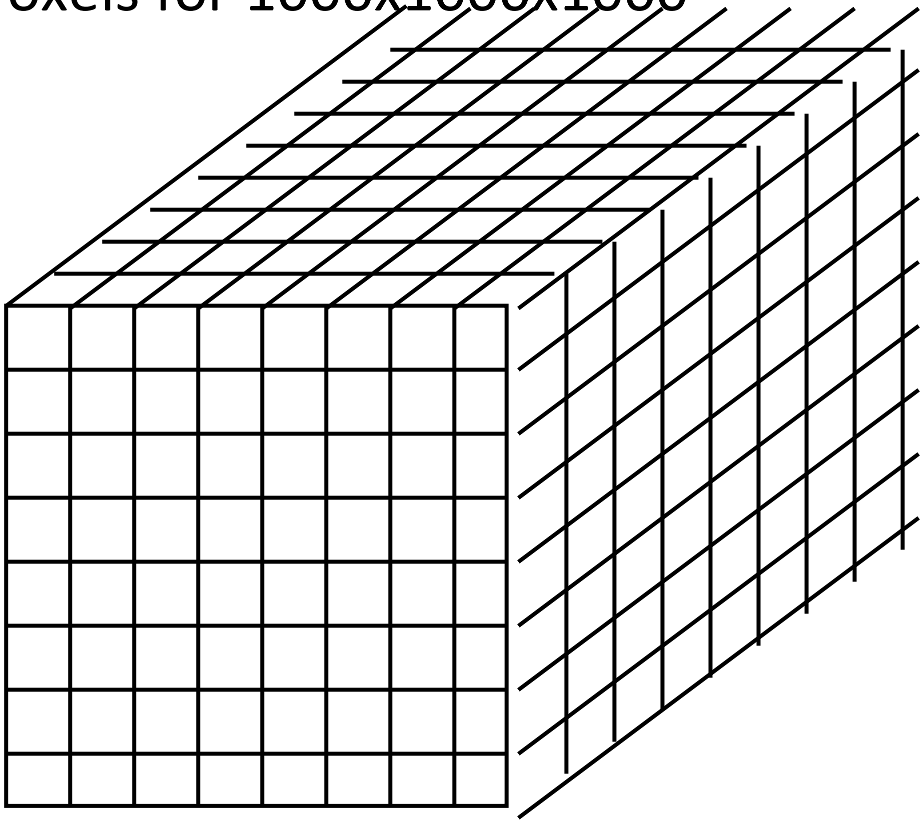
- Partition space into uniform grid
  - Grid cells are called a *voxels* (like pixels)
- Store properties of solid object with each voxel
  - Occupancy
  - Color
  - Density
  - etc.



FvDFH Figure 12.20

Voxel Storage

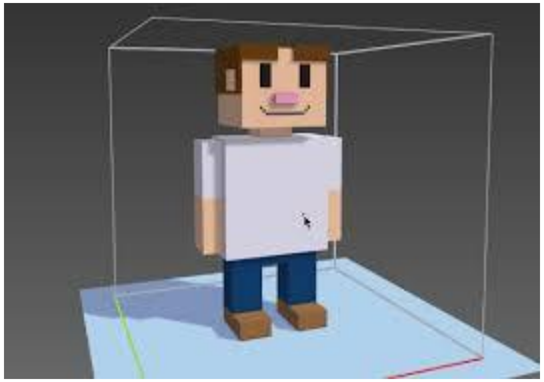
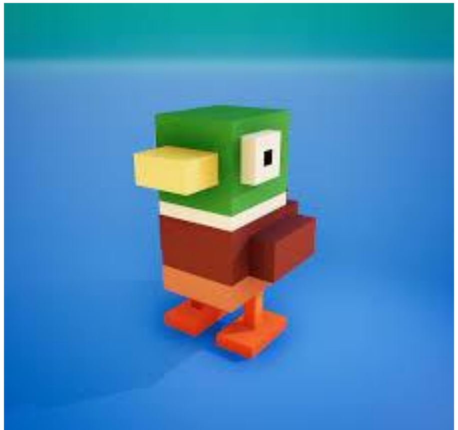
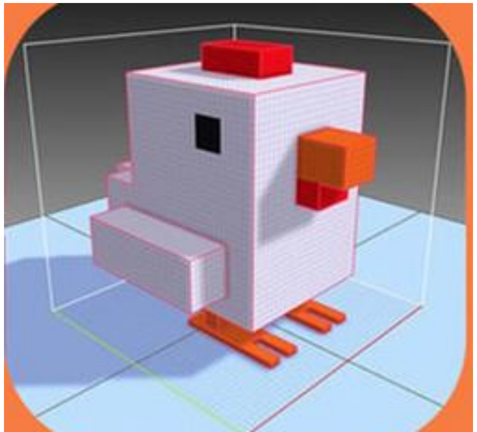
- $O(n^3)$ storage for NxNxN grid
    - 1 billion voxels for 1000x1000x1000



Voxel Art Examples

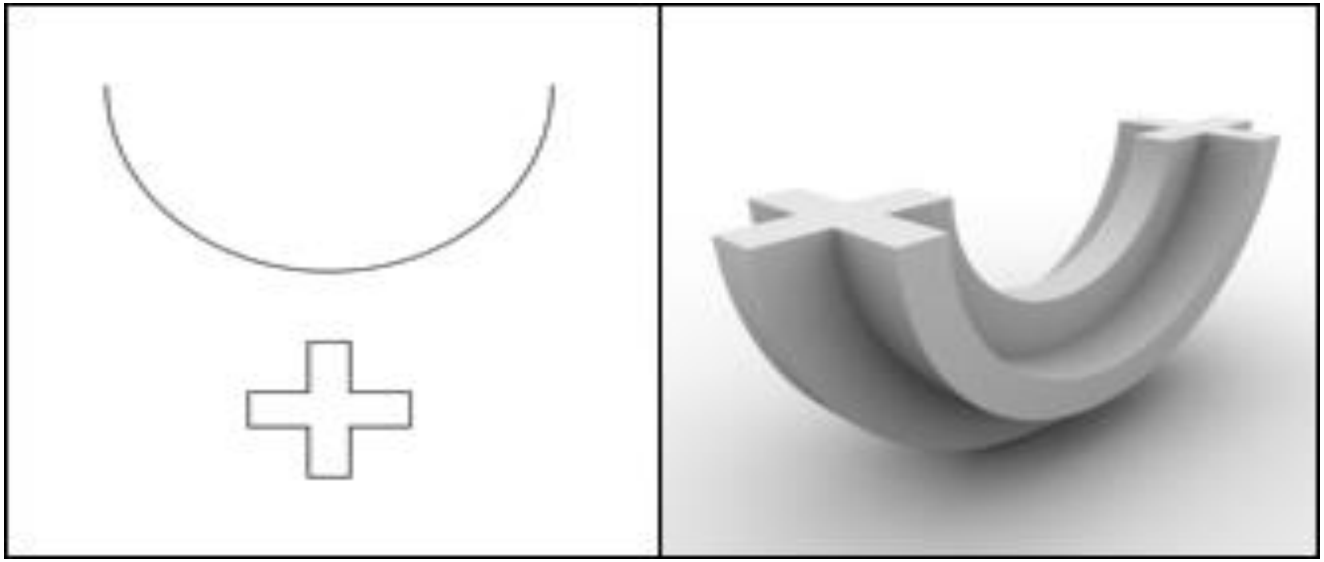
- Using Voxels to Build Simple Shapes with low quality
  - Example Tool (MagicaVoxel: https://ephtracy.github.io/)



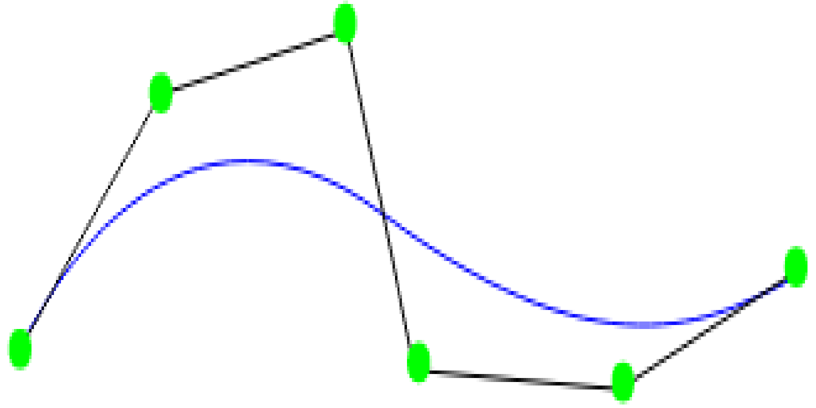
9-NURBS modeling

- NURBS stands for **Non-Uniform Rational B-spline**.
- In NURBS modelling, lines and surfaces are not manipulated by moving vertices, edges, faces or polygons.
- Instead NURBS surfaces and lines are manipulated by **special control points**. The following 2 techniques make use of NURBS.

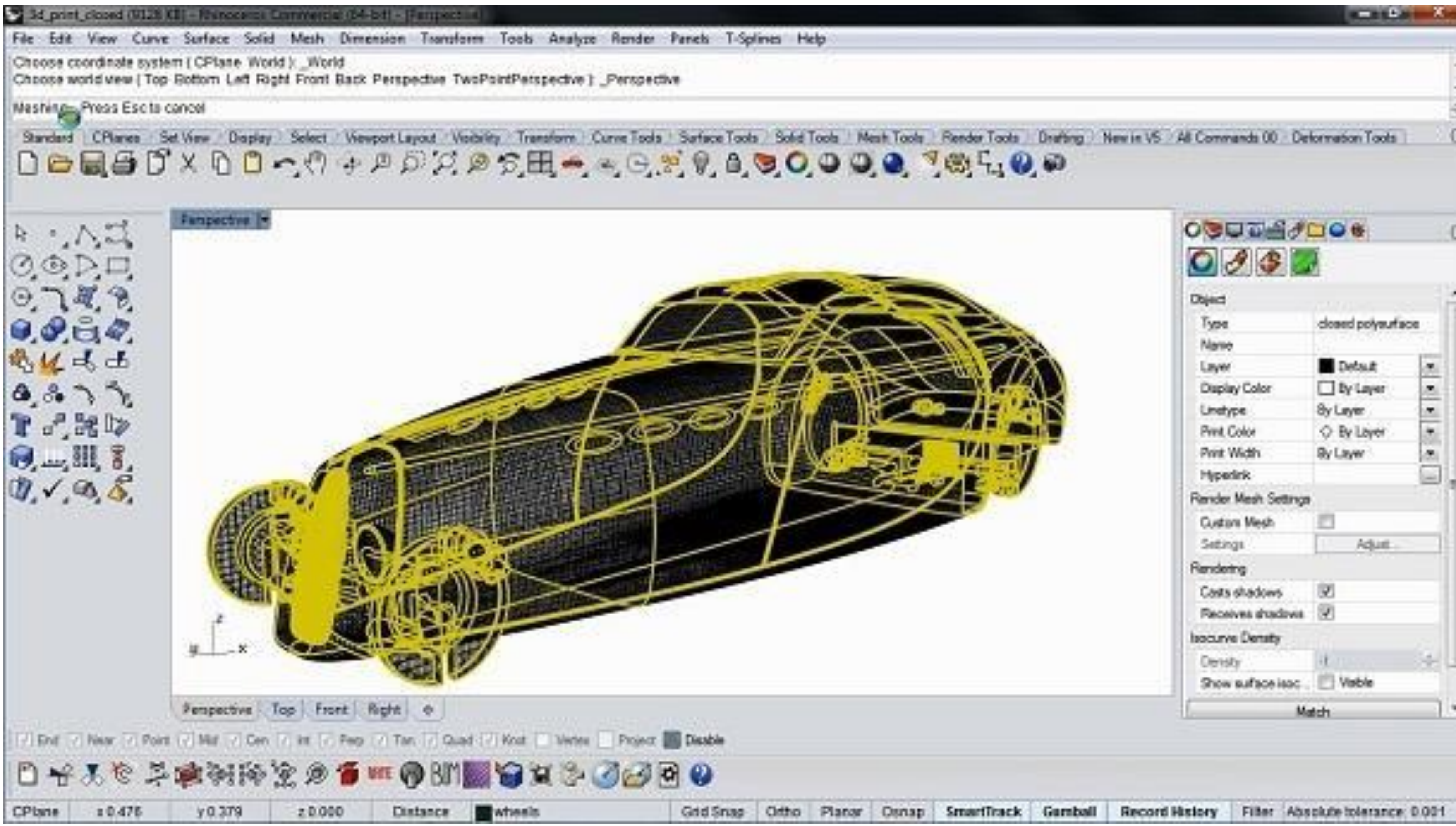


NURBS Control points

- A NURBS model is a mathematical modeling type commonly used to generate **curves** and surfaces. The main advantages of this modeling technique are the great flexibility and precision you have in generating your shapes.
- In contrast to **Polygon Modeling** (which we will discuss a little bit later), curves are created with a tool that works very similarly to the pen tool in MS paint or Adobe Illustrator. The curve is drawn in a 3D space, and edited by moving a series of handles called **CVs** or **control vertices** (see below).



NURBS Example



Projection types

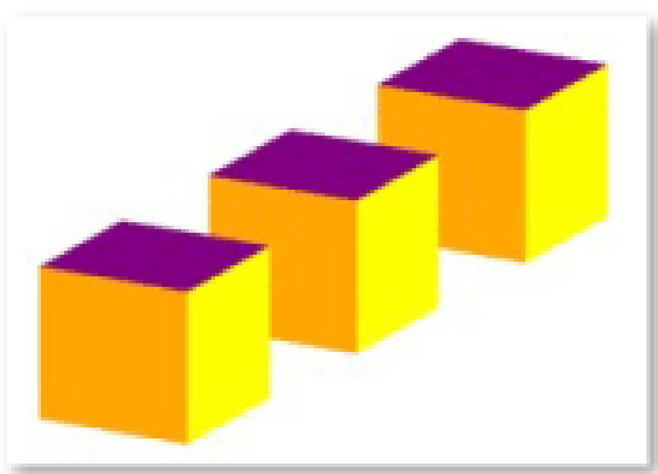
Projection transformation: then defines the camera settings. It sets up what can be seen by the camera. It contains 2 types: perspective and orthographic.

1. **Perspective View:** the configuration includes field of view, aspect ratio and optional near and far planes.

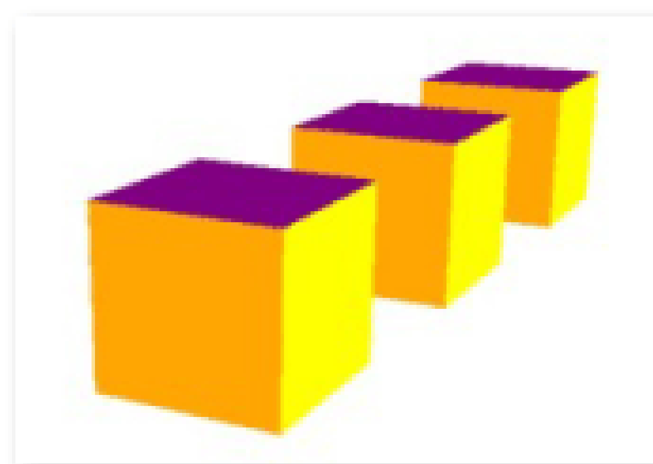
   A perspective camera is how we see the real world. If we take a look at the things around us, they have depth and we can judge their distance.

2. **Orthographic view:** Objects are drawn without perspective distortion.

Orthographic Projection

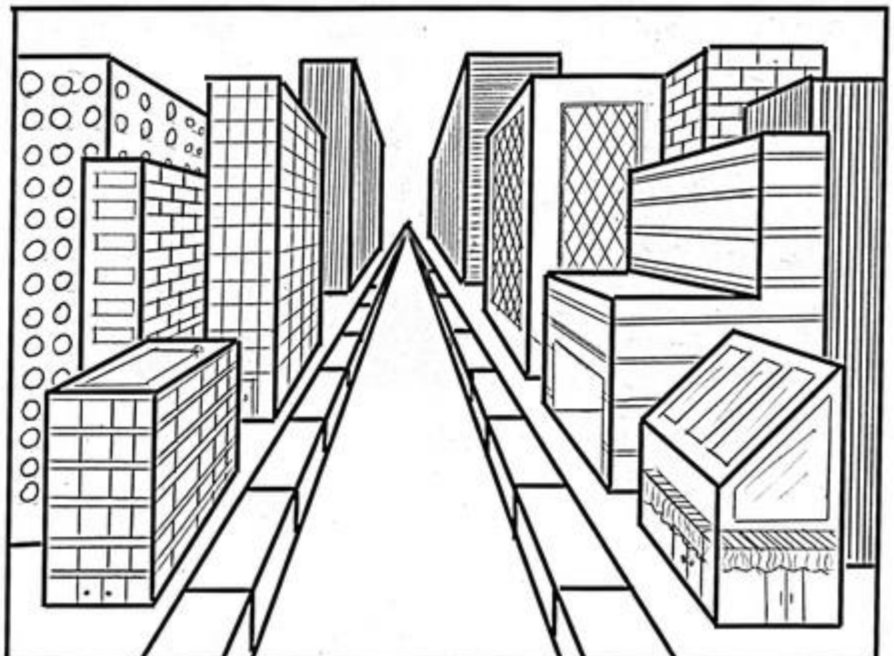


Perspective Projection

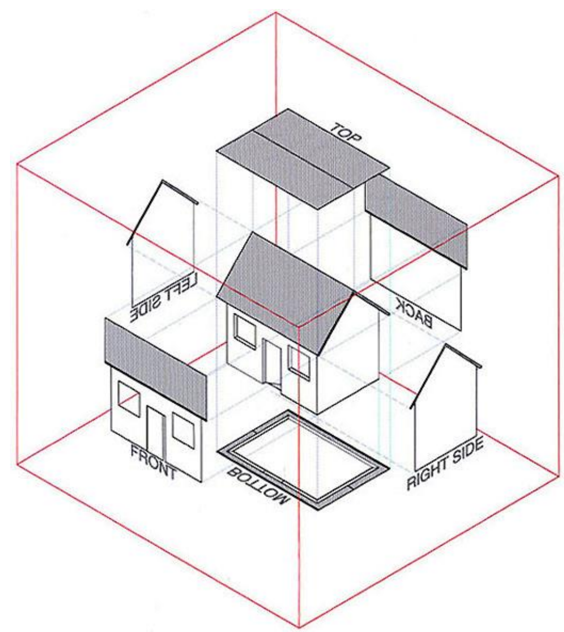


Projection Types

- Perspective projection
  - Just like our eyes and cameras

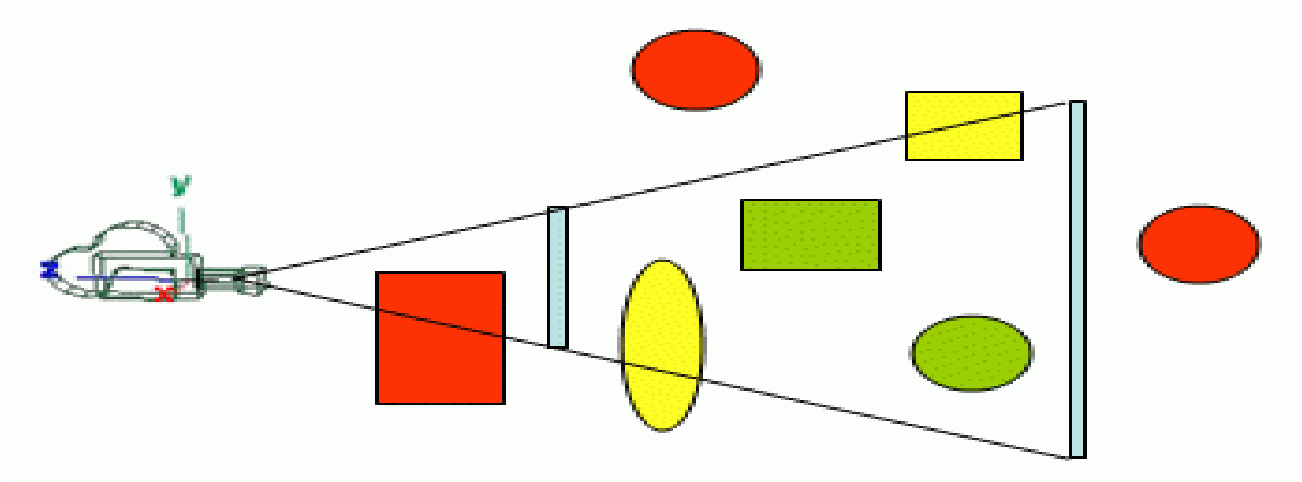


- Orthographic projection
  - Architectural drawing with no distance distortion



View Frustum
الشكل المخروطي للمشاهدة

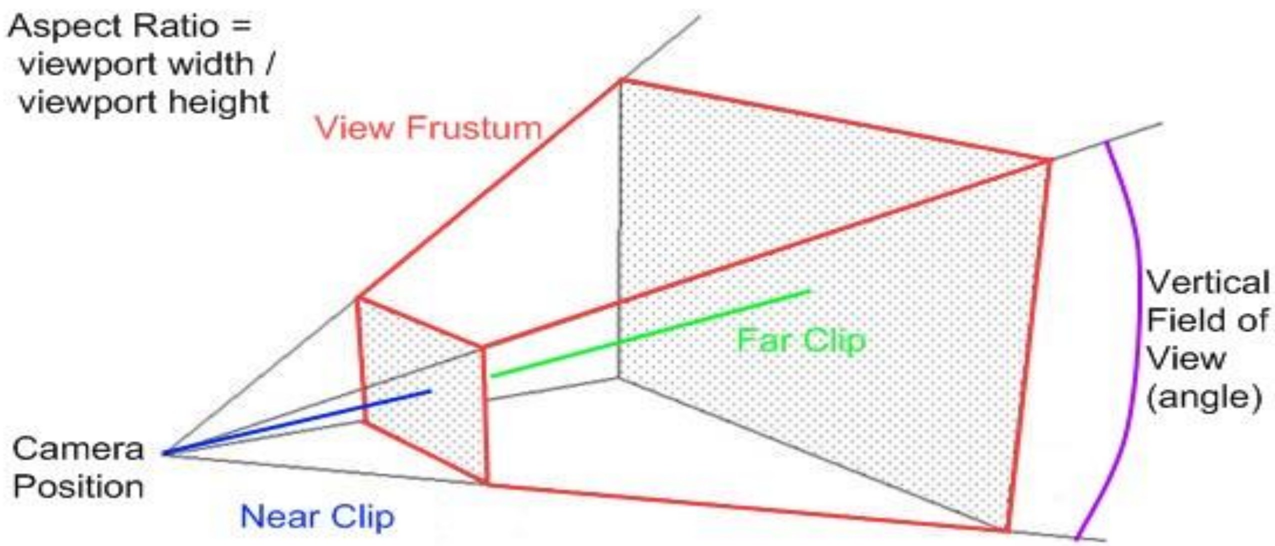
- The view frustum is the **volume** that contains everything that is **potentially** (there may be occlusions) **visible** on the screen.
- This volume is defined according to the camera's settings, and when using a perspective projection takes the shape of a truncated **pyramid**.



https://www.cnblogs.com/linuxheik/articles/10882425.html

View Frustum Elements

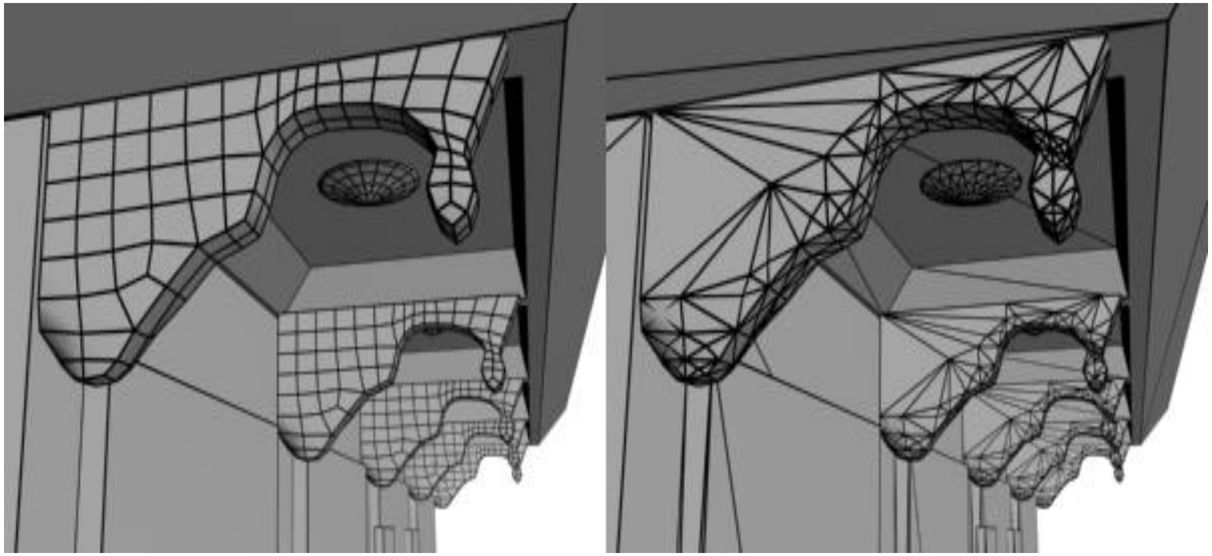
- **Top, Left, Right and Bottom**: they basically make up the sides of the frustum.
- **Near plane:** This stops objects from coming too close to the camera. So if anything come closer than the near plane, it gets chopped off.
- **Far Plane:** This sets how far the camera can see. Objects that are far away and are too small to see properly will still be processed and draw, which is a waste so this limits, is a way to limit the max distance for the camera to improve efficiency.
- **Field of View (fov)** is an angle inside the view frustum camera holds which describes how much the camera can see in terms of angles. If field of view changes, the displayed image on the screen will change. If fov decreases (e.g. from 90 to 45), you will be able to see less on your left and right of the world and it looks like you have zoomed in (like when you use sniper rifles in fps games). Whereas if you increase the fov (e.g. from 70 to 100), you can see more of the world and it looks like you have zoomed out compared to lower fov.



https://ksgamedev.wordpress.com/2010/01/11/week-5-camera-and-spaces-part-2/

Topology

تركيب الشكل الداخلي



In 3D modeling, the term *topology* refers to a 3D model’s edge distribution and structure.

Two models that look the same when rendered can have very different topologies.

The model on the left is constructed mostly with quads (four-sided polygons) while the model on the right is constructed with triangles (three-sided polygons), also known as tris. These models are said to have different topologies.

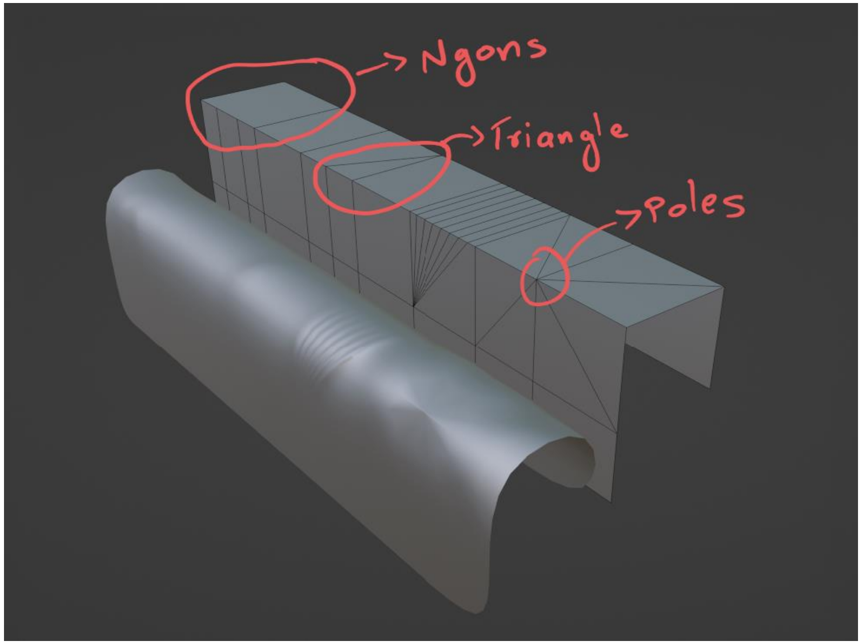
Topology affects many aspects of 3D modeling and rendering, including:

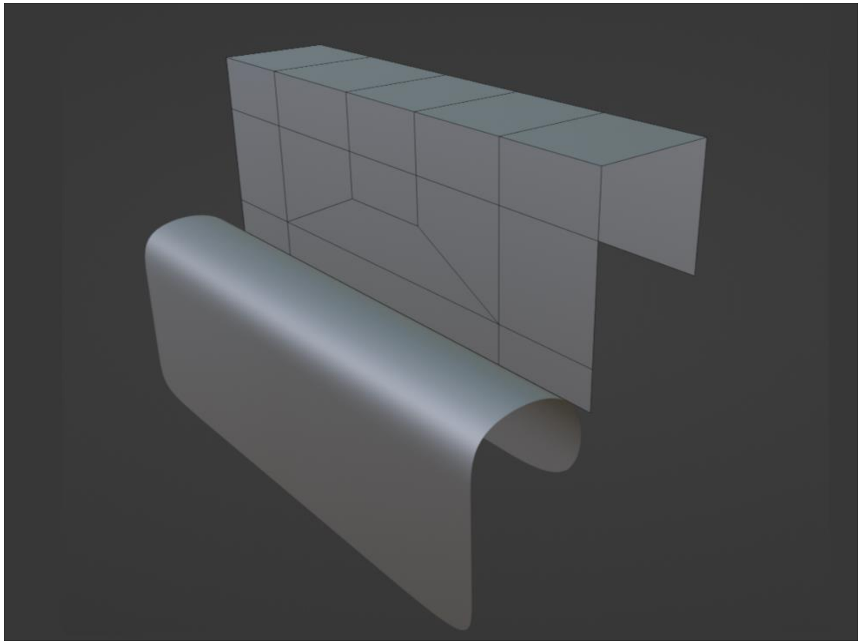
- How highlights react to the model
- How easy it is to edit the model’s shape
- How easy it is to apply or edit mapping. Unwrapped UVs are easier to manage with quad topology.
- How easy it is to rig and animate the model. Quad topology makes it much easier to rig and animate.

https://resources.turbosquid.com/checkmate/checkmate-product-presentation/topology/

Problems with Non-Quad Topology

- Let's model a simple mesh with bad topology and deform it. Then add a subdivision surface modifier to the mesh.
- As you can see, weird artefacts are seen with topologies containing **Ngons**, **triangles**, and **poles**. These artefacts increase with the angle between faces.
- A **pole** is a set of edges that merge into a single vertex. Avoiding poles with six or more edges on a curved surfaces is something that you should incorporate into your modeling workflow.
- The target is to make Topology (Clean) (almost quad)
- Note: the Right Figure is the same shape as the Left Figure with a cleaner topology

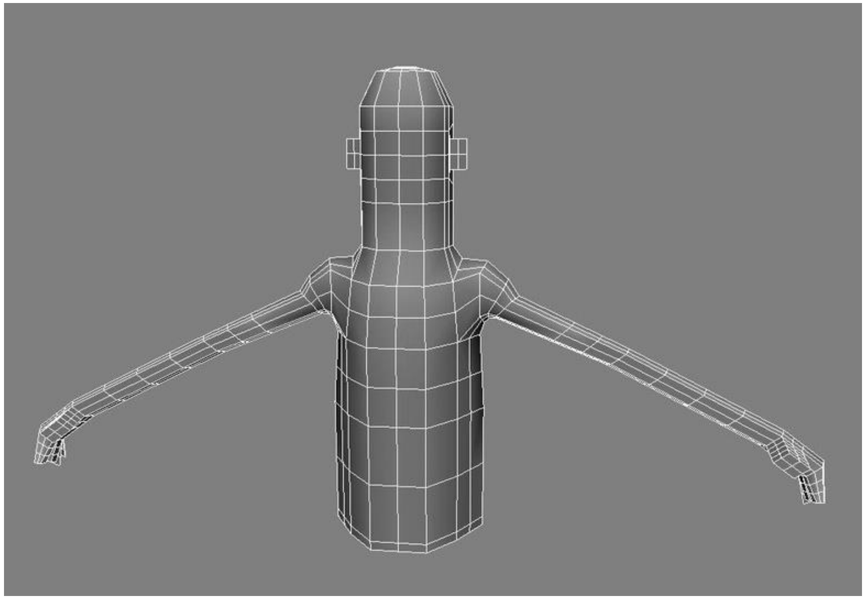




https://blog.yarsalabs.com/blender-topology-guide/ , https://blog.yarsalabs.com/blender-topology-guide/

Basic Clean Topology Tips

1. Use four-sided (Quads) polygons freely
2. Avoid N-gons that have five or more sides.
3. Use Less triangles sparingly as possible
4. Let quads relatively square.

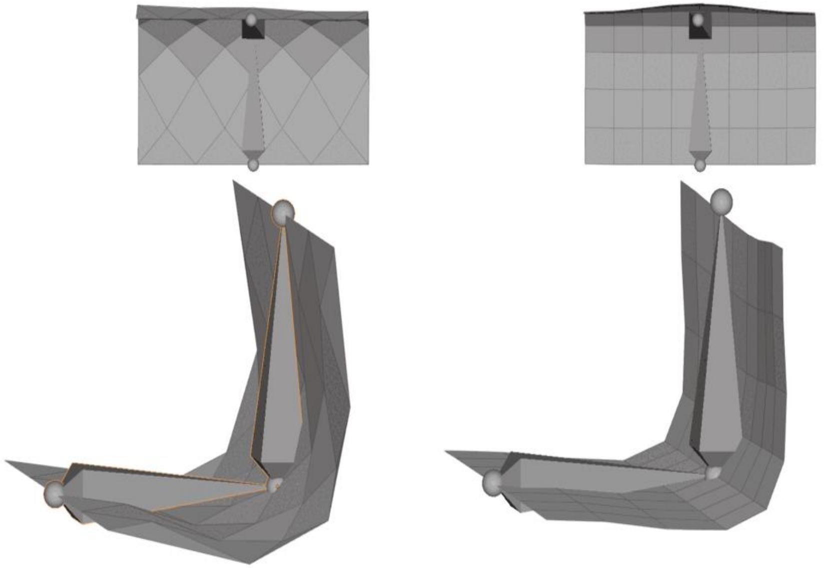


https://blog.yarsalabs.com/blender-topology-guide/ , https://blog.yarsalabs.com/blender-topology-guide/

Clean Topology for Animation

Align Faces with the Axis of rotation (along the Edges)

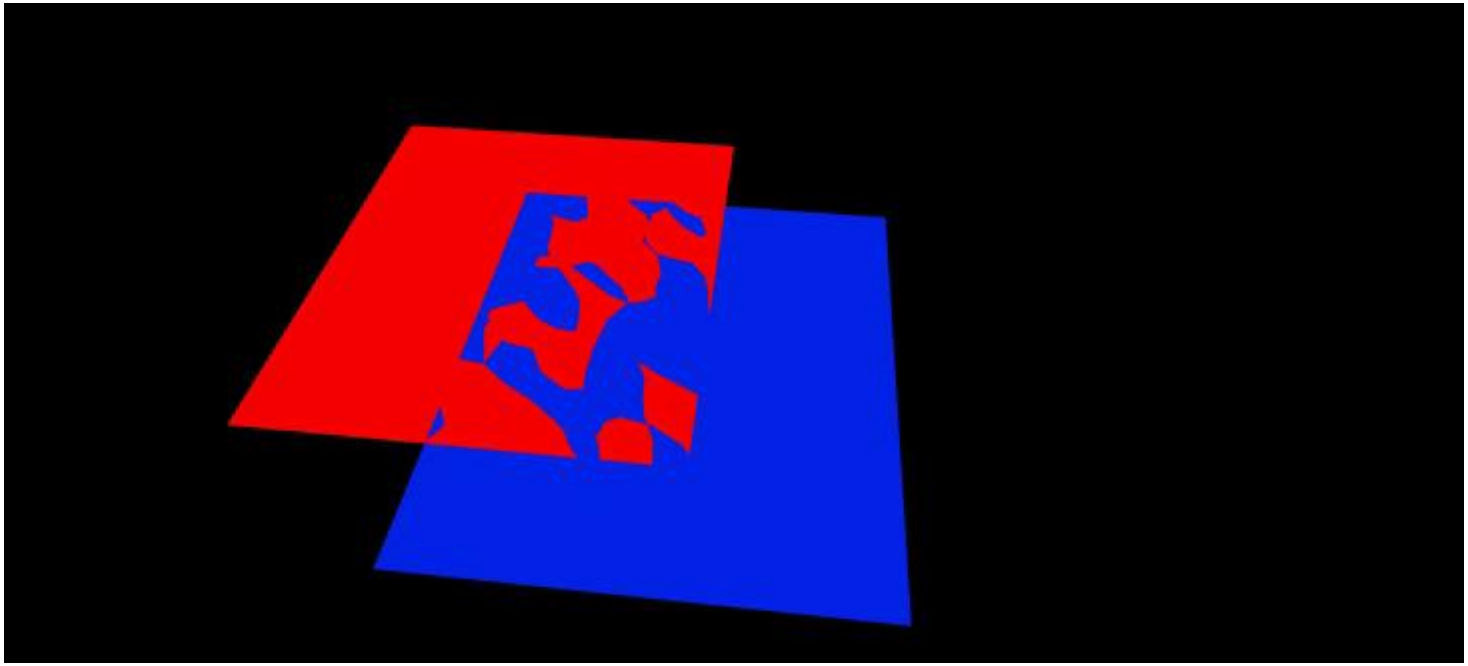
Note in the left shape: the rotation of corner triangles are not bending correctly



https://topologyguides.com/

What is Z-Fighting?

- Z-Fighting is where multiple objects fight to be rendered closest to the camera *(fighting to be on top)*. This can happen when the objects are overlapping due to having similar z-buffer values as the scene is rendered.

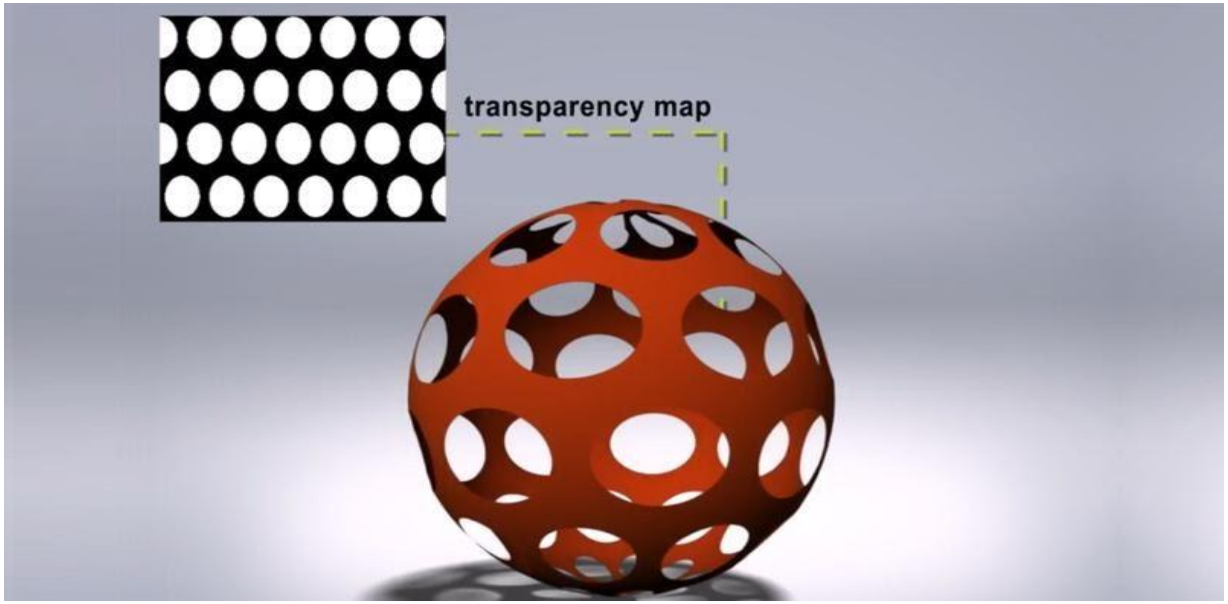


Textures
صورة لتغليف الشكل

- Textures are 2D images used in the 3D space to make the objects look better and more realistic. Textures are combined from single texture elements called texels the same way picture elements are combined from pixels. Applying textures onto objects during the fragment processing stage of the rendering pipeline allows us to adjust it by wrapping and filtering it if necessary.

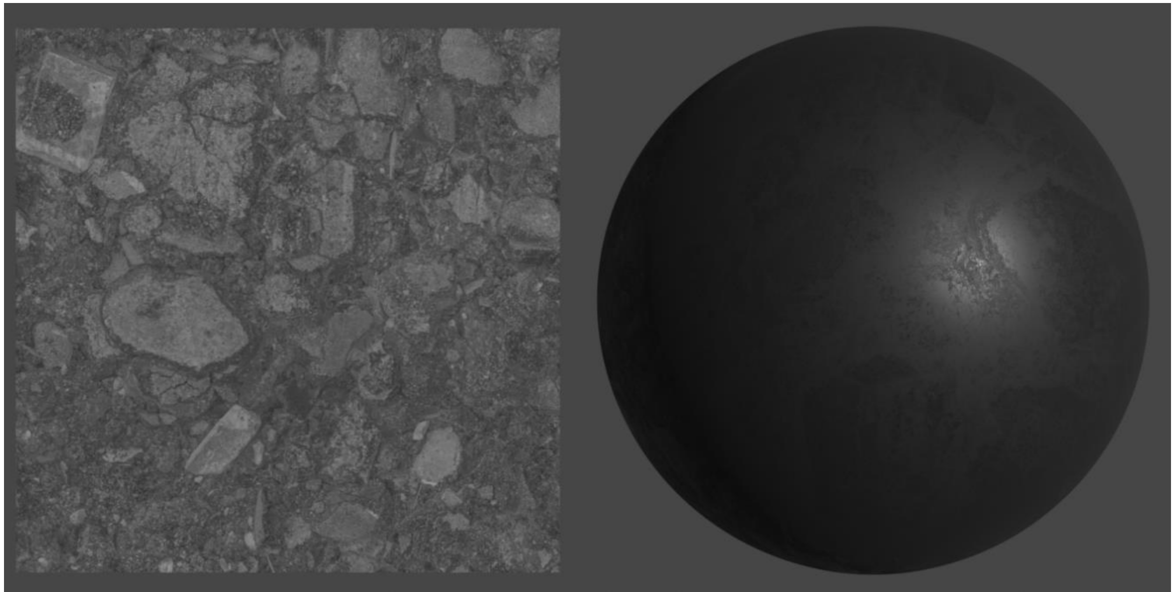
Types of texture Maps-p1

1. **Color map**, **diffuse** map, or **Albedo** map: Gives the object **colours**
2. **Metalness** map: Tells you the metallic معدني properties of the material.
3. **Roughness** map or gloss map: Tells you the glossiness لمعان of the surface, determines how reflective the material is.
4. **Transparency Map:** grey scale textures that use black and white values to signify areas of transparency or opacity on an object's material



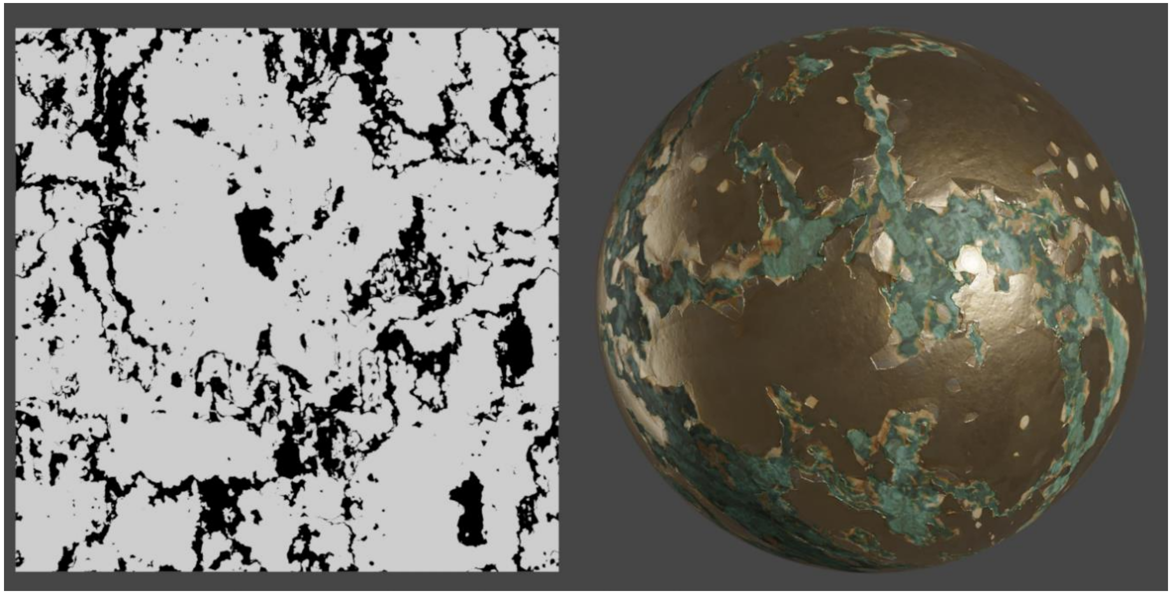
Gloss (or Roughness)
لمعان(أو خشونة)

- The terms **Gloss** and **Roughness** are interchangeable - they are simply the inverts of each other. كلمتان متعاكستان
- So if your software calls it "Gloss" like (Marmoset), but there is only a "Roughness" map provided, simply invert it in the software (usually a checkbox is provided) before connecting.
- It often adds a lot of character to hard surface materials like wood or metal



Metalness

- When downloading one of our Metalness workflow materials, which we provide for all our metal materials. You'll find an extra map include called Metalness.
- It tells the renderer what part of a surface should be treated as Metal and what parts should be treated as non-metal. Metals reflect light very differently to everything else so this map can make a huge difference to the final look.
- [Ref: https://help.poliigon.com/en/articles/1712652-what-are-the-different-texture-maps-for](https://help.poliigon.com/en/articles/1712652-what-are-the-different-texture-maps-for)

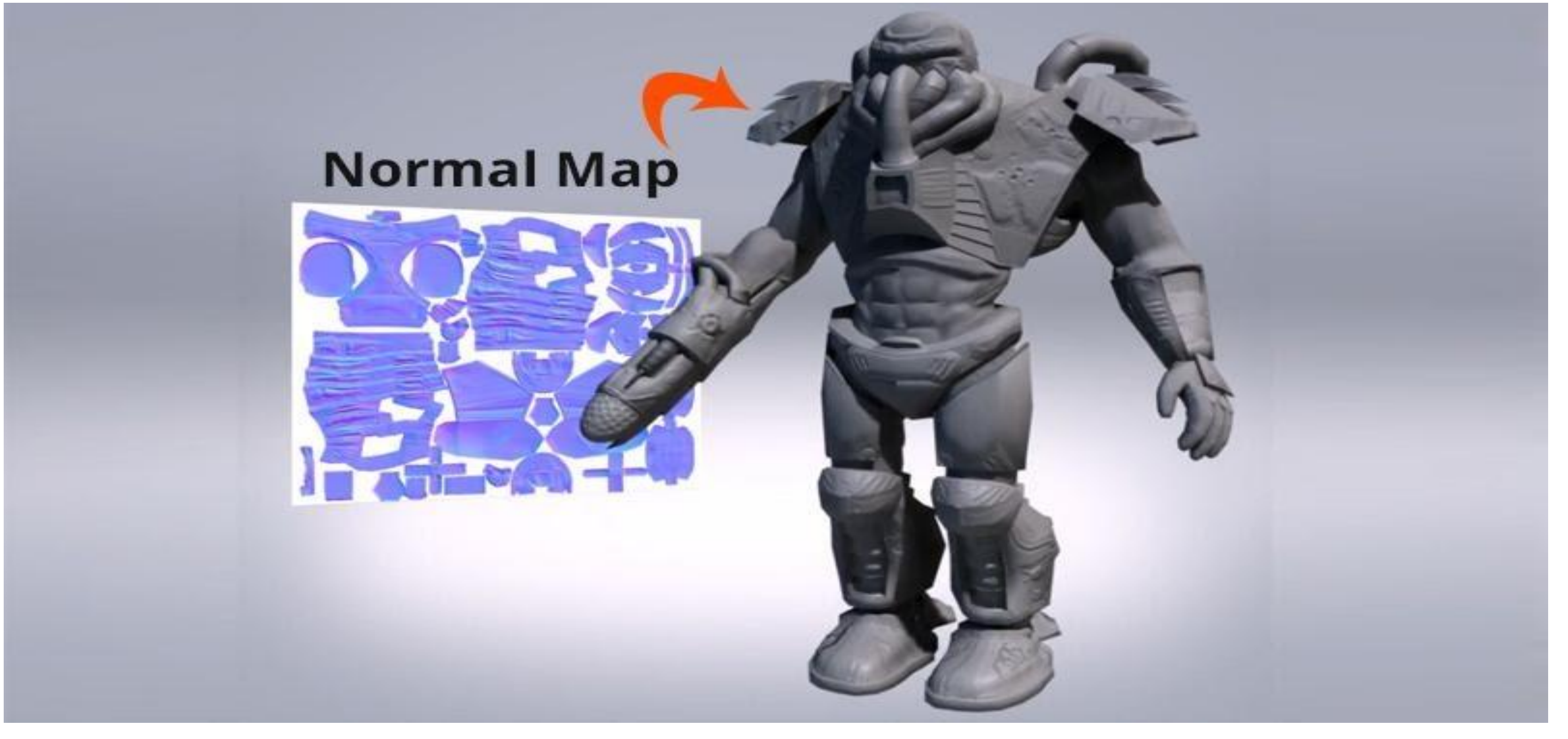


Types of texture Maps-p2

5. **Bump** map: Creates the illusion of depth by faking اصطناع بدون تغيير فعلي details on the surface using height information to tell when a point should be up or down. The surface however is still flat and not actually warped.

Types of texture Maps-p3

6- Normal map: A type of bump map that fakes the details by using angle information to tell which direction each point of the surface should be oriented towards. This gives better effect with lighting and shadows.



Types of texture Maps-p4

7- Displacement map: Similar to a height map but actually changes the geometry of the object by moving vertices (points) on the mesh (surface). This physically alters the form of the object.

Normal Maps VS. Bump Maps

- both affect the normals of your geometry and create the illusion of detail without having to rely on extra geometry
- **bump maps** just encode height information using black and white values
- **Black**: minimum height delta
- **White**: maximum height delta
- **normal maps** use RGB values to signify the orientation of the surface normals:
- red, green and blue channels = X, Y and Z orientation of the surface
- Normal maps can typically get **more detailed** information onto the surface

PBR Textures

- **PBR, or Physically Based Rendering, is a model of shading used in computer graphics that more closely mimics the flow of light in the real world.**
- PBR textures produce a higher quality of realness with greater variations of lighting

https://cgaxis.com/how-to-use-pbr-textures/

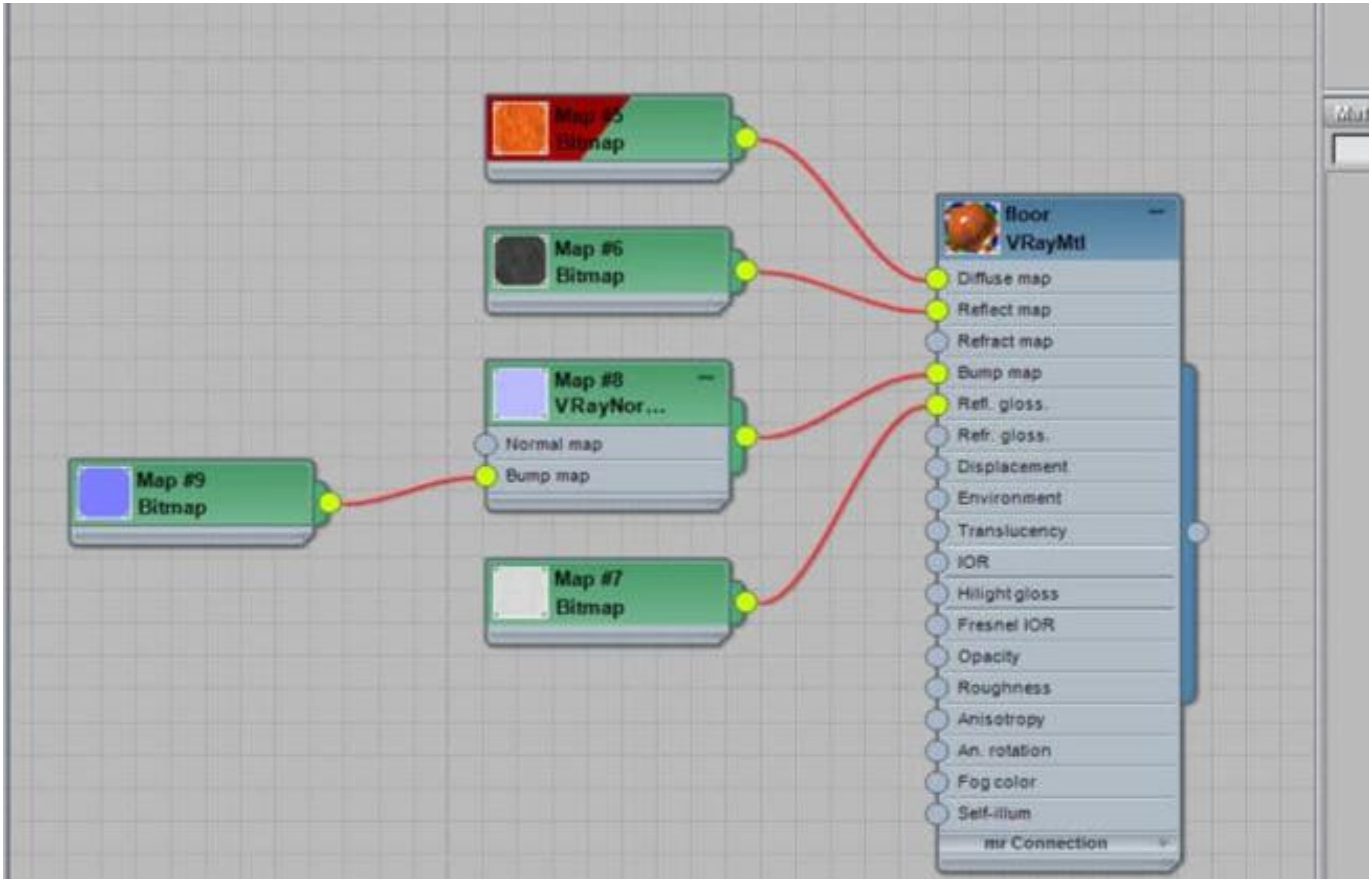
PBR Textures Components

To create material in specular workflow you will need the following maps:

1. **Albedo (Diffuse) Map** – base color input, which is the most important map, giving information about the color of the object.
2. **Specular Map** – defines an object's shininess لمعان by using a greyscale map. In its practical application the brighter the shade, the more light will be reflected.
3. **Glossiness Map** – this map controls the reflectivity لون انعكاس الضوء of an object, whether the reflection of light will be blurred or sharp.
4. **Normal Map** – this map determines the number of dents and bumps an object has. Similar to a bump map yet more advanced, the normal map is saved in RGB format, containing its information in XYZ axes.
5. **Height** (**Displacement**) Map – this map is saved as a grayscale which creates a larger group of convexities and creases which affect the objects mesh.

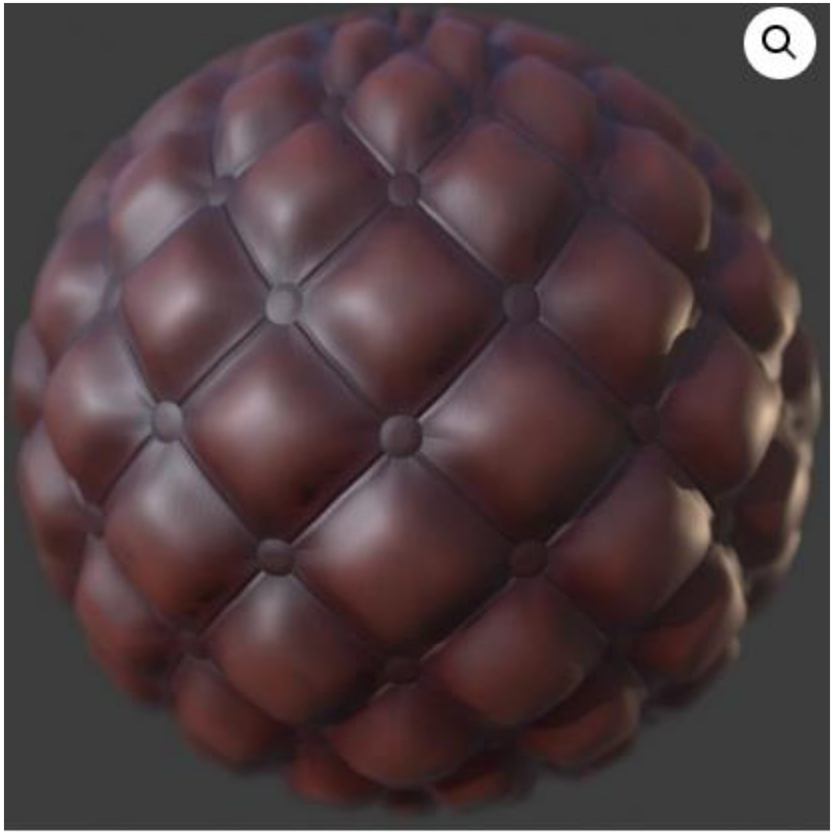
https://cgaxis.com/how-to-use-pbr-textures/

PBR Editor Example (3D max V-Ray)



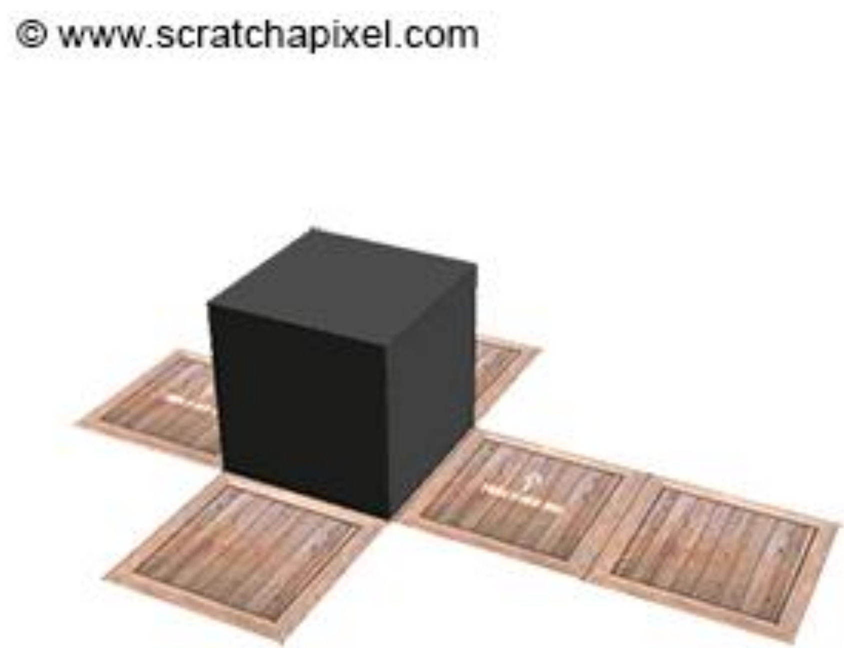
Free PBR websites

- https://freepbr.com/
- List of free websites:
- https://thegraphicassembly.com/best-websites-for-free-pbr-textures/

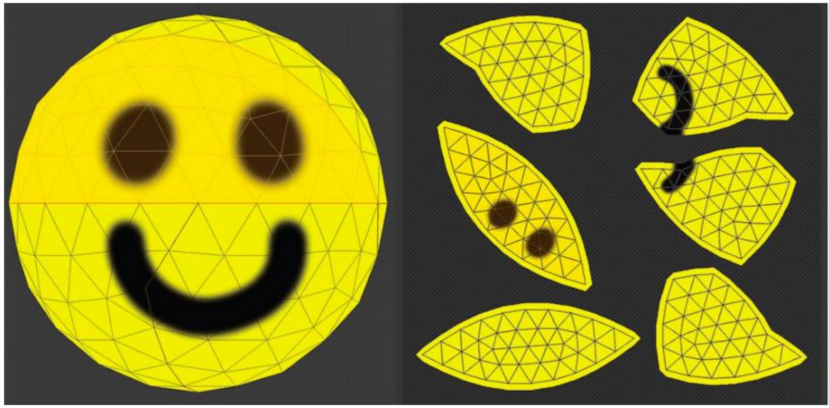


Texture Coordinates (UV mapping)

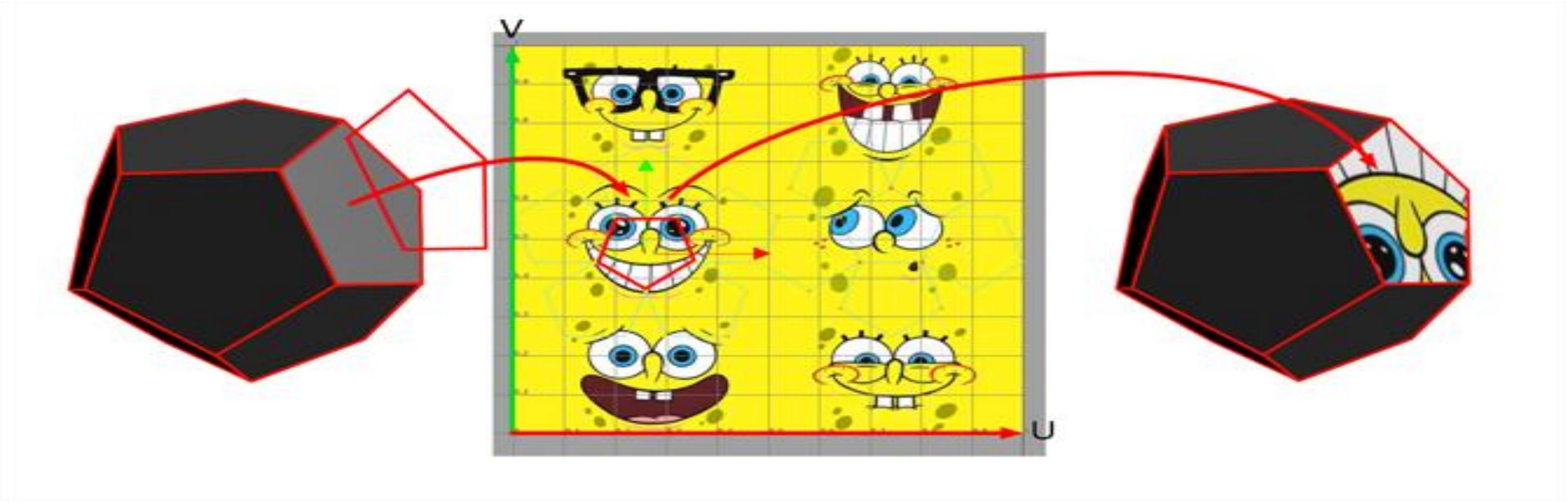
- commonly referred to as UV mapping. You have a model, and a texture that you want to apply to it. The texture has various areas on it, representing images that we want to apply to different parts of the model. There has to be a way to mark which triangle should be represented with which part of the texture. That’s where texture mapping comes in.



UV-mapping

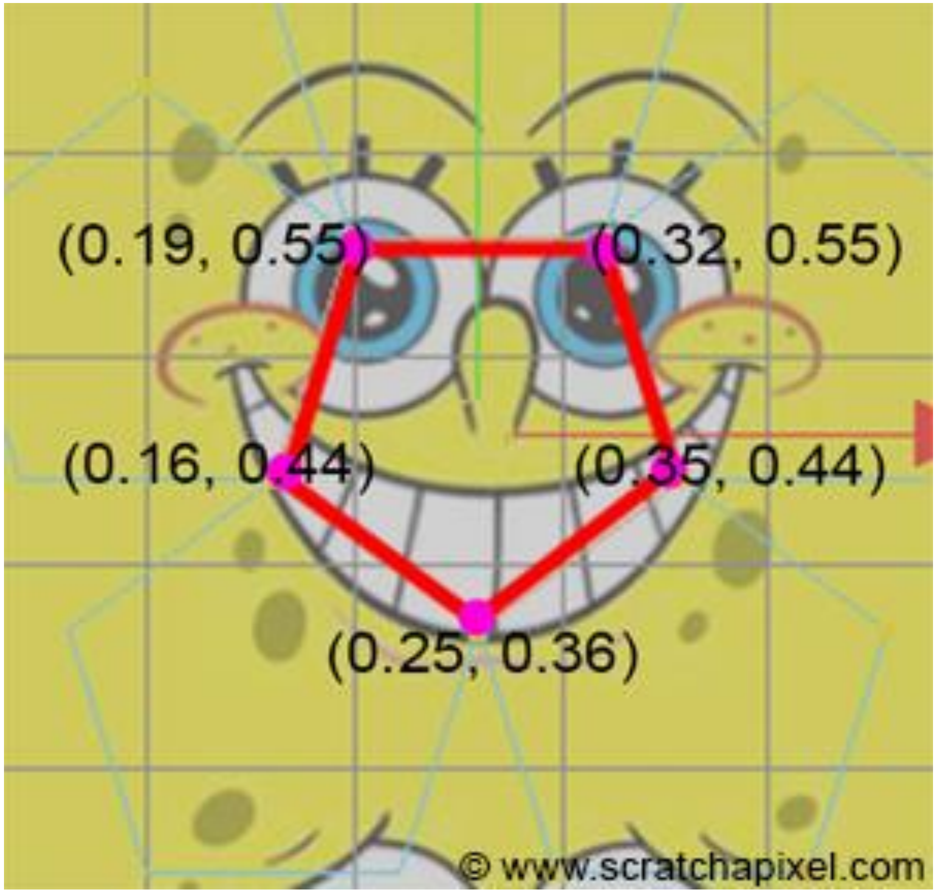


- For each vertex, we mark two coordinates, U and V. These coordinates represent a position on the texture, with U representing the horizontal axis, and V the vertical axis. The values aren't in pixels, but a percentage position within the image. The bottom-left corner of the image is represented with two zeros, while the top-right is represented with two ones.
- A triangle is just painted by taking the UV coordinates of each vertex in the triangle, and applying the image that is captured between those coordinates on the texture.



UV-mapping--example

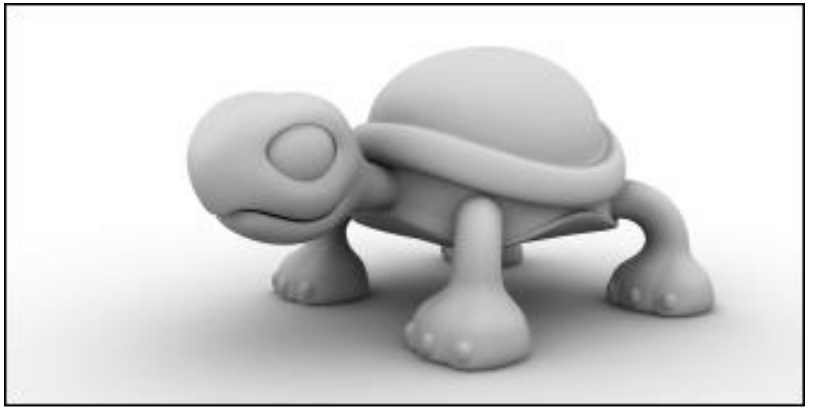
- Note : U and V values range from 0 to 1



Rendering types

1-Clay Model Rendering::

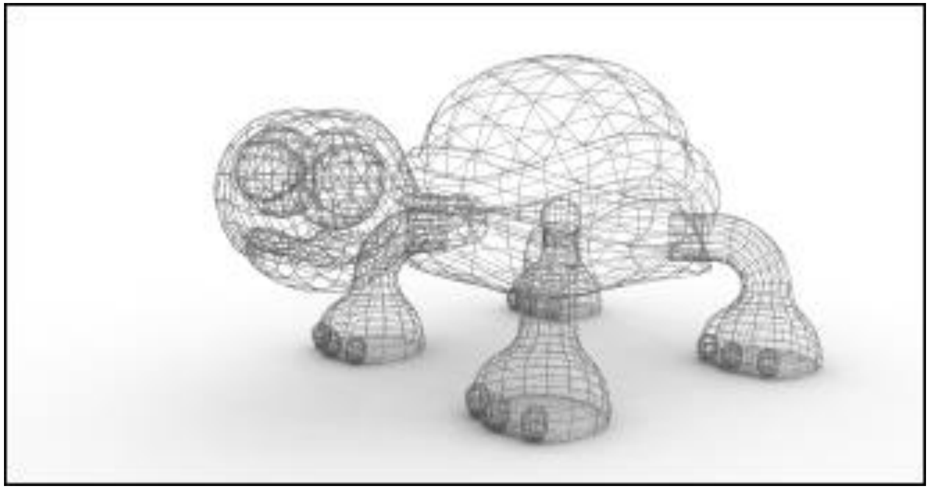
- When 3D modeler's present their modelling skills they often produce a clay render. Clay render means a rendering which looks like a picture of a clay model.
- This is not really a rendered effect however as it is generally the default form of a non textured model
- Useful for design



Rendering types

2-Wireframe Rendering::

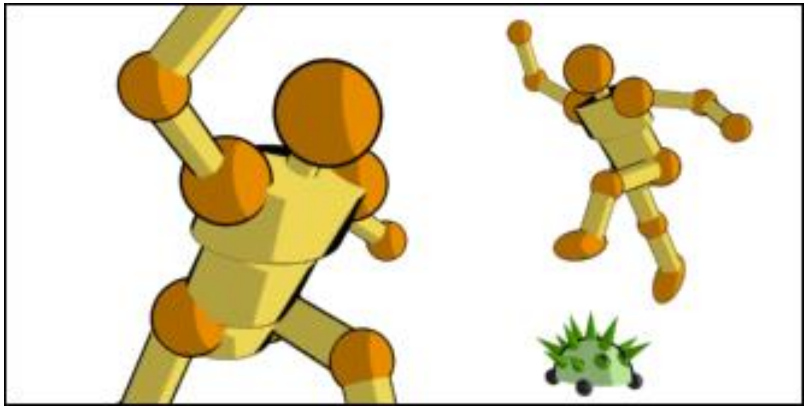
The purpose of wire rendering is to reveal the polygon structure of the model. Wire rendering displays only edges of the polygons. Wire render and clay render are often combined.



Rendering types

3-Cell Shading Rendering::

Cell Shading is a from of Non-photo realistic rendering (NPR). In contrast to traditional computer graphics, which has focused on photo realism, NPR is inspired by artistic styles such as painting, drawing, technical illustration, and animated cartoons



Rendering types

4-**Photorealistic Rendering**

- **Photorealistic Rendering** is a modern technique that renders the images more realistic and clearer than ever.

https://it-s.com/what-is-photorealistic-rendering

3D animation Phases

1-Modelling the character:

to create the basic model

2- Rigging the character:

to add bones

3-Skinning the character:

to make a relation between model parts and bones

4-Animating the character

Rigging the character

- Character rigging means a process of creating the bone structure for a character.

Standard Humanoid Bone systems-1

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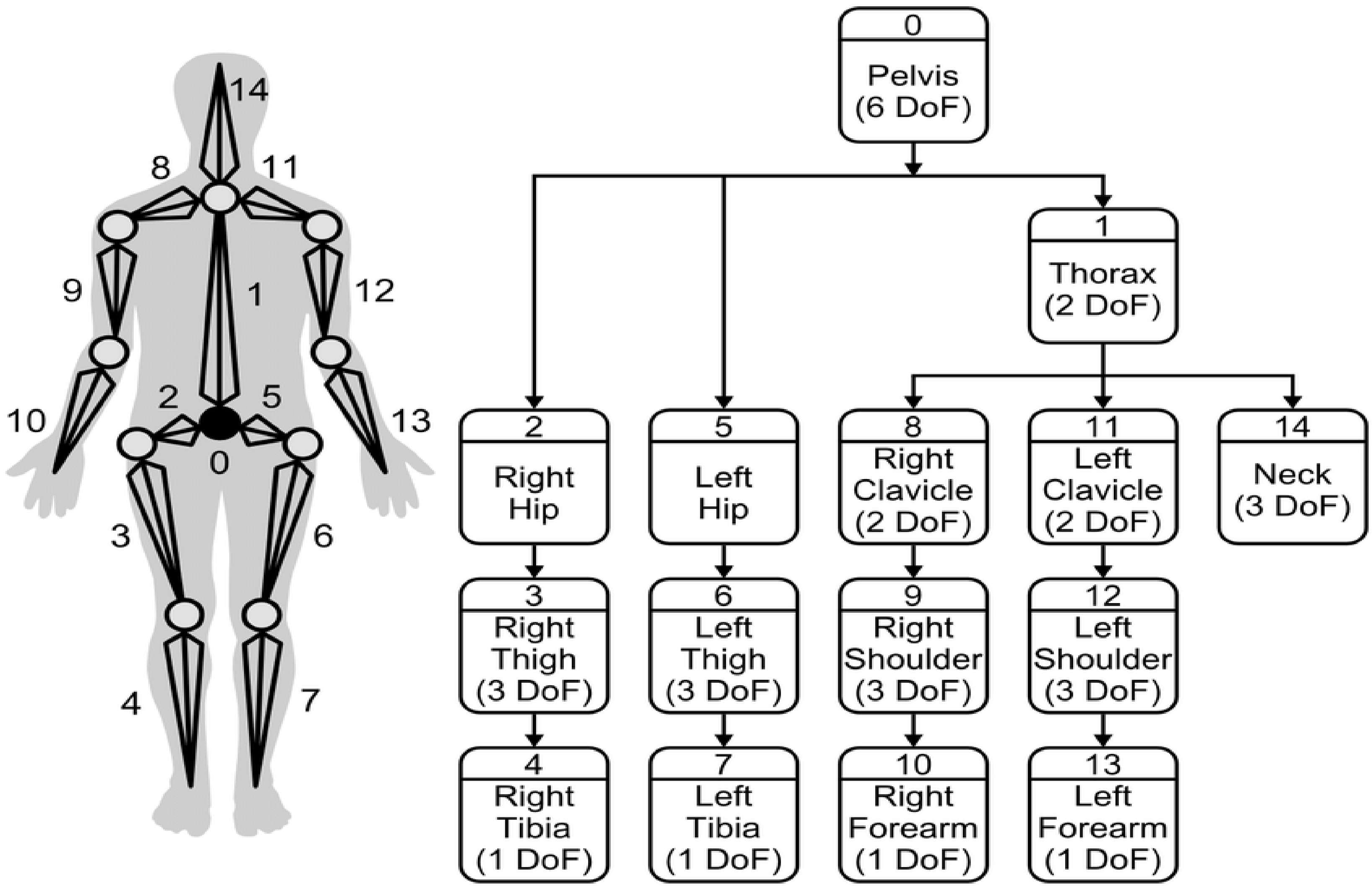
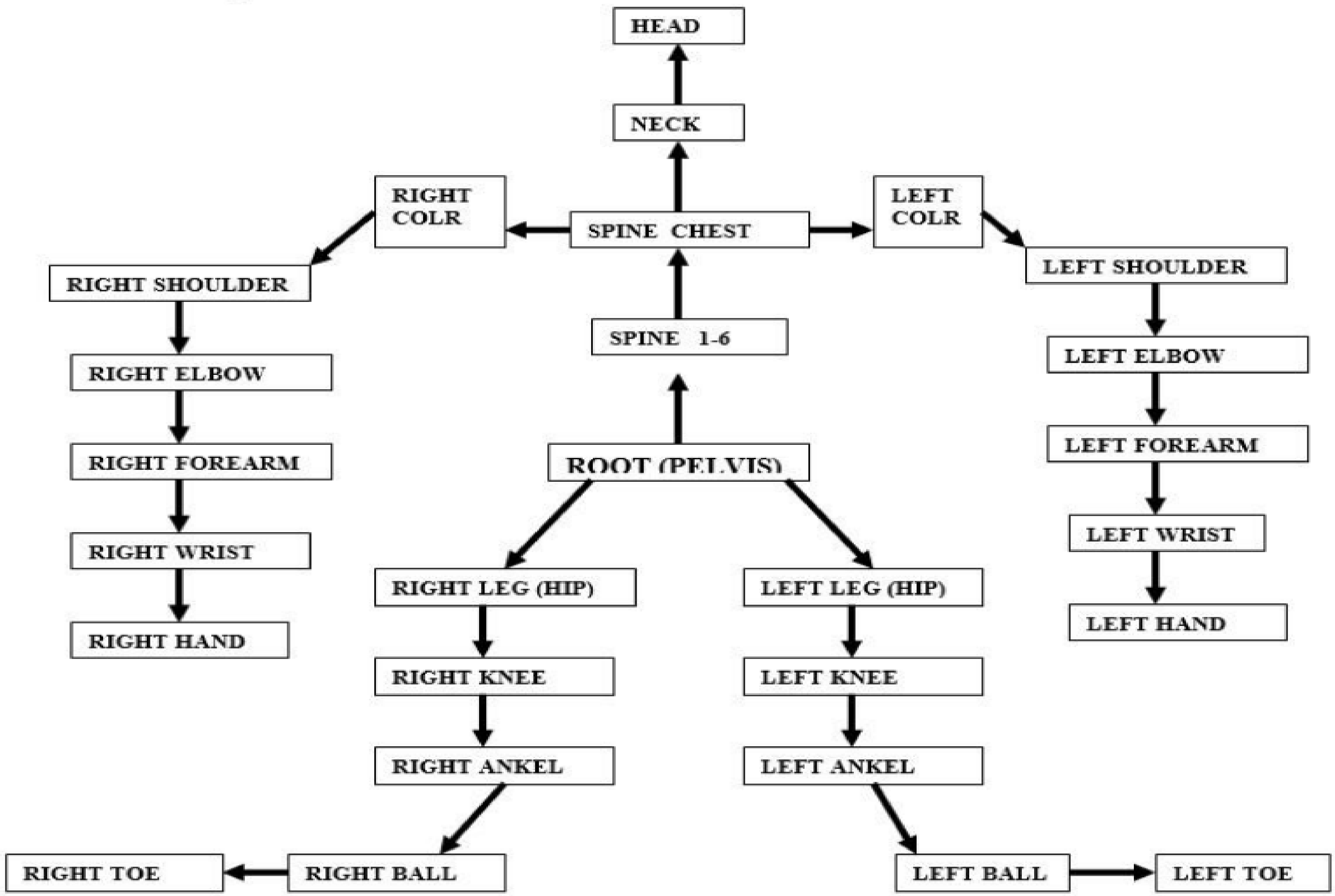


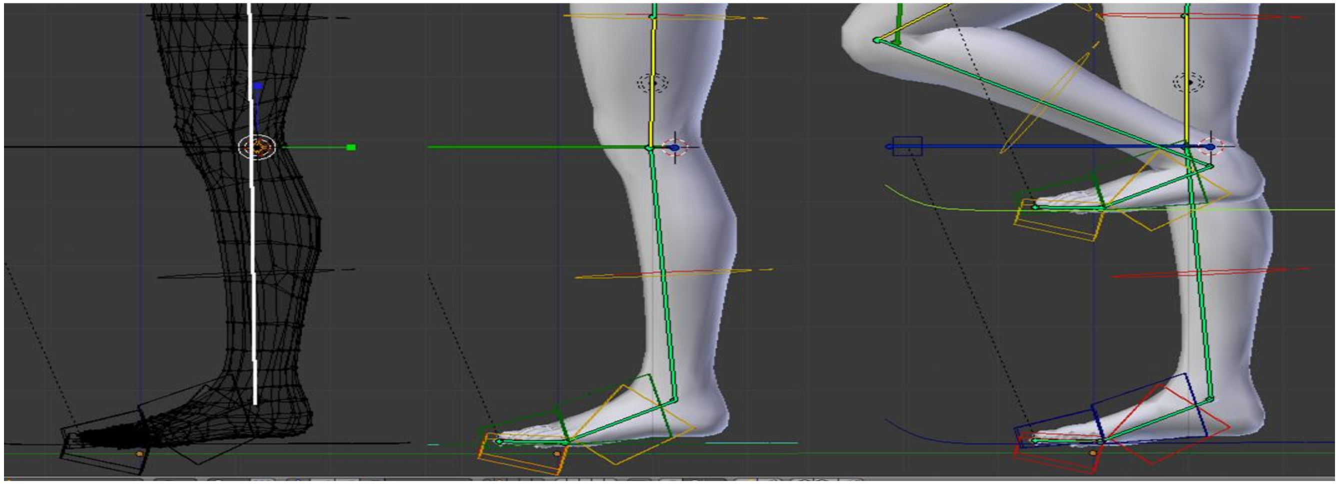
Standard Humanoid Bone systems-2

للاطلاع فقط



Skinning the character

- Character skinning means the process of defining how exactly the character responds to the movement of the bones. In the skinning process one goes through all the joints in the bone structure and carefully adjusts how the 3D model deforms while a certain bone is moving. As an addition to bones there are also other special tools and modifiers designed to help in character animation. For example facial animation is often carried out by morphing the face between several predefined states.



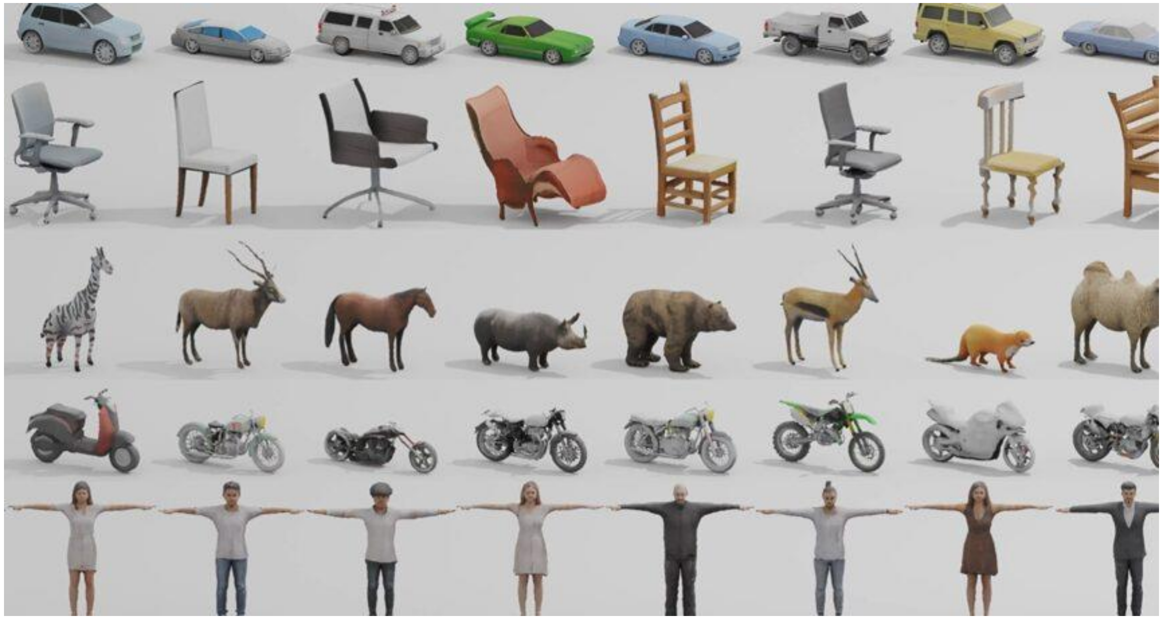
Animating the character

- Animating requires a lot of skill and practice. An animator should understand at least the basics of character movement such as the concept of anticipation. Character animation process can be fluid when rigging and skinning are done with care.

Part 2- Research Trends and Topics

1- Generate 3D models from 2D images or videos

This allows for rapid prototyping and the creation of realistic models from real-world data.

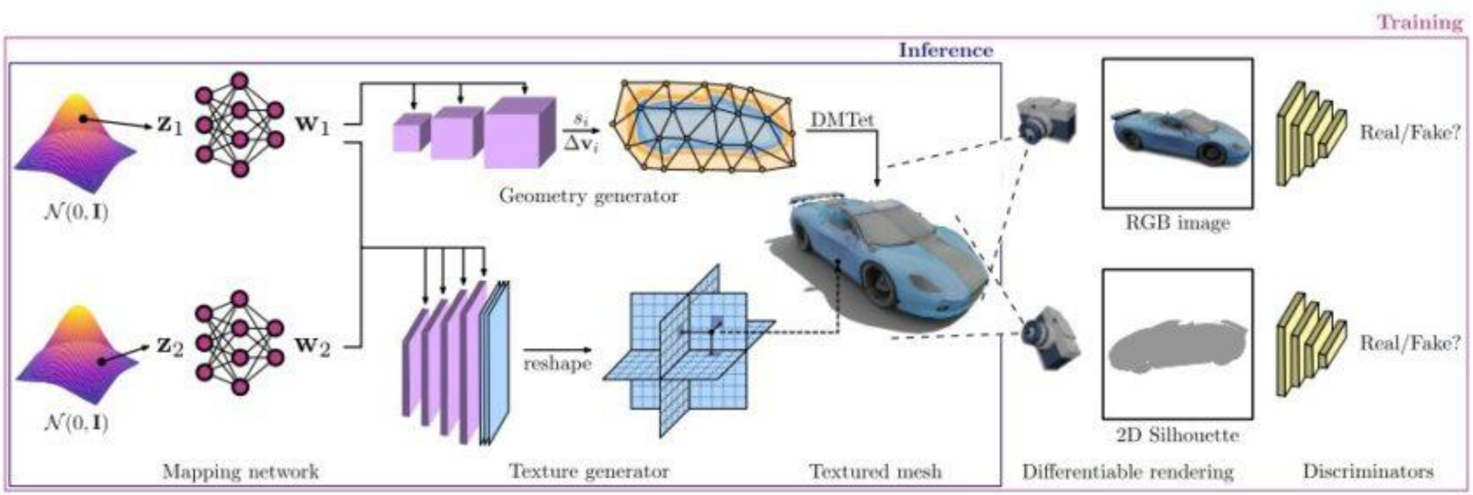


https://mixed-news.com/en/nvidias-latest-open-source-ai-generates-3d-models-from-a-single-2d-image/

Example : Nvidia's 3D model generation from synthetic 2D images للاطلاع فقط

Nvidia's research team has developed a two-step generation process: The geometry branch generates the polygon mesh with any desired topology. The texture branch generates a texture field that can represent colors and, for example, specific materials at the surface points of the polygon mesh.

Finally, as with GA networks, discriminators evaluate the quality of the output based on synthetic photos of the 3D model and continuously optimize it to match the target image.

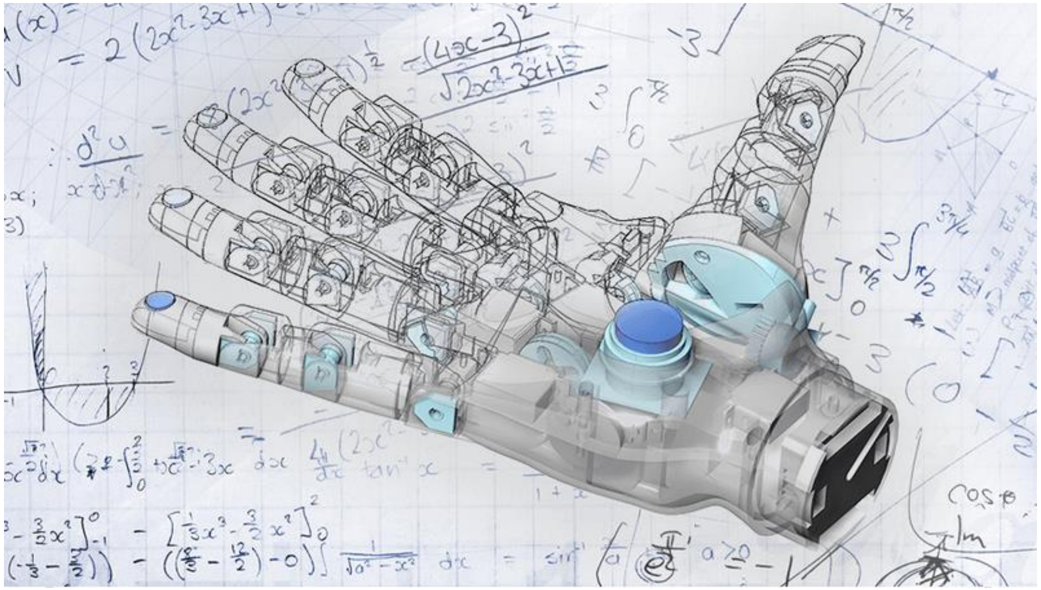


GET3D

https://github.com/nv-tlabs/GET3D

2- Optimize 3D models for specific purposes

AI can analyze a model and suggest changes to improve its performance, such as reducing its size or complexity.

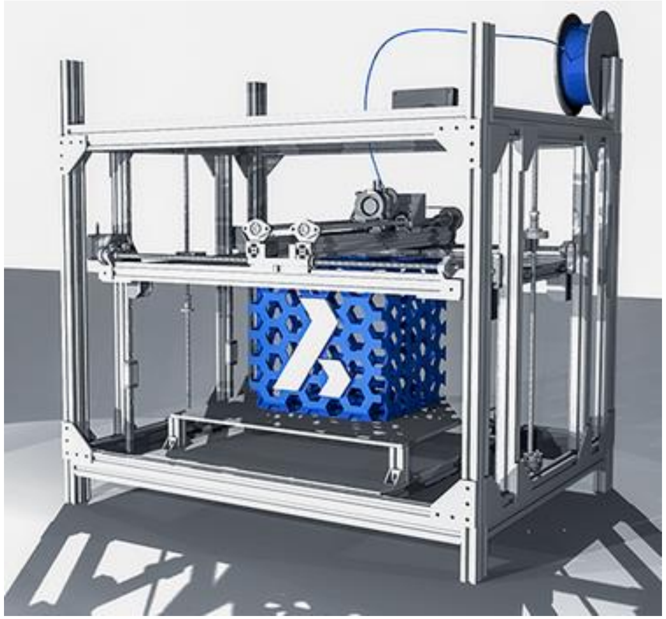


3- Use AI instead of traditional CAD methods

the traditional CAD process is well established. But current market advances have brought about significant threats to its success. AI-powered generative 3D design is paving the way for the future of CAD technology.

Generative design, AI and technological advances such as VR facilitate a flexible design process that maximizes creativity and empowers designers to do their best work.

In general, it looks something more like this:



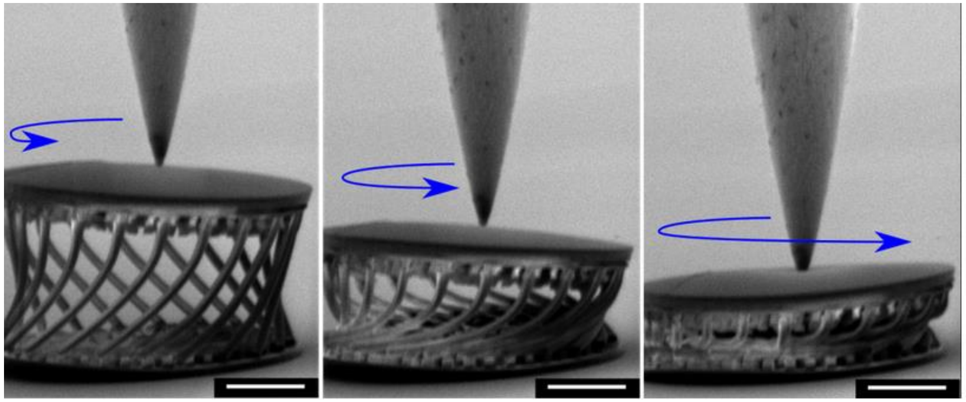
4- Enhance 3D Printing

Printing 3D designs makes it possible to produce product prototypes at incredible speed. Artificial intelligence removes geometrical limitations from design options. This lets you make the most out of your 3D printing technologies.

3D printing is now making it possible to manufacture more and more complex generative designs. The two in tandem result in design flexibility and heightened product functionality.

When generative design and 3D printing align, the results can be truly exceptional. Some of these improvements include:

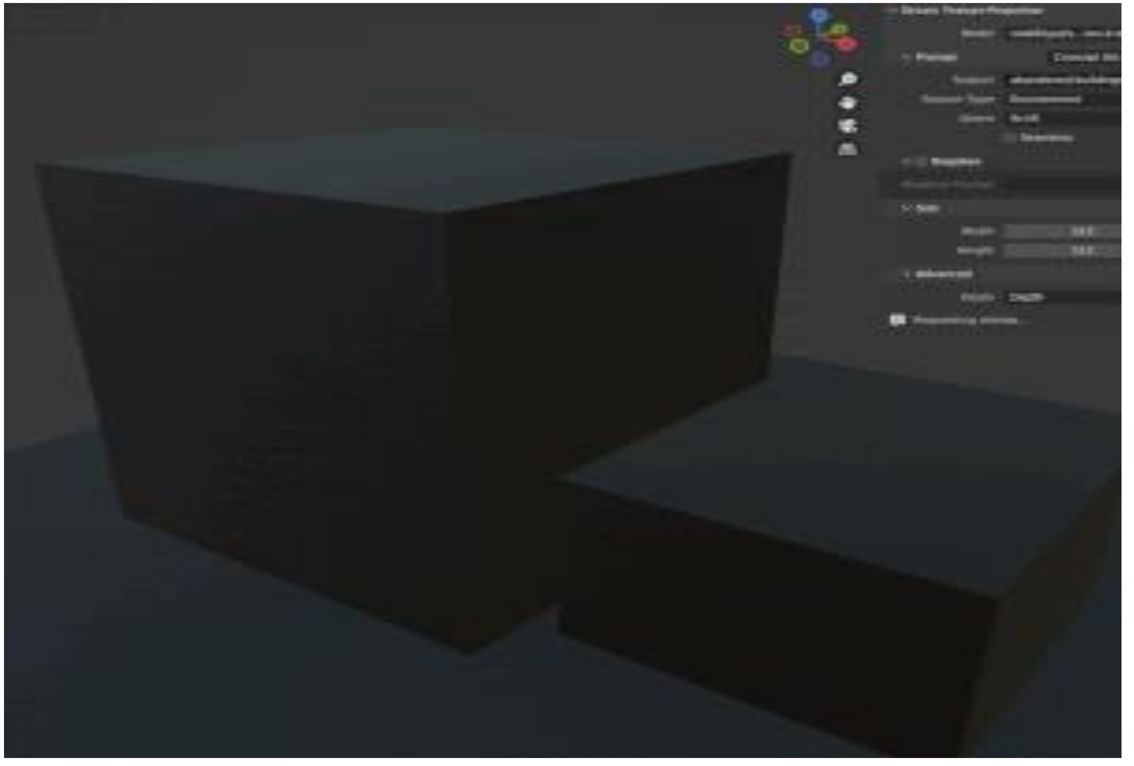
- Lightweight parts
- Less material wastage
- Efficient prototyping and testing
- No geometrical boundaries
- Innovative design options
- Consolidated parts
- Simplified assembly
- Reduced costs



Nanoscale version of the AI designed material under compression. Image with special thanks to Miguel Bessa and T U Delft.

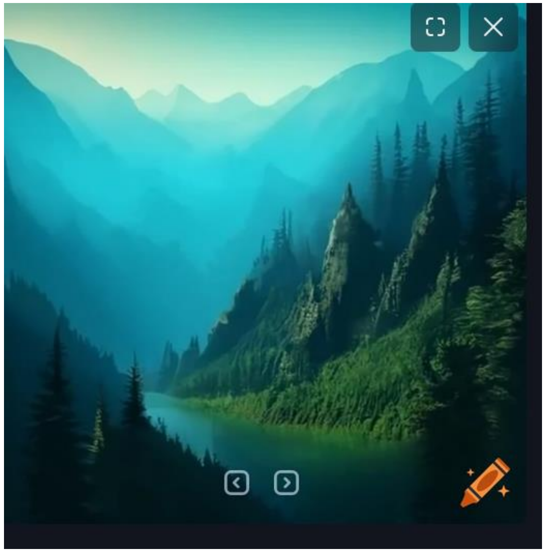
5- Create new textures and materials

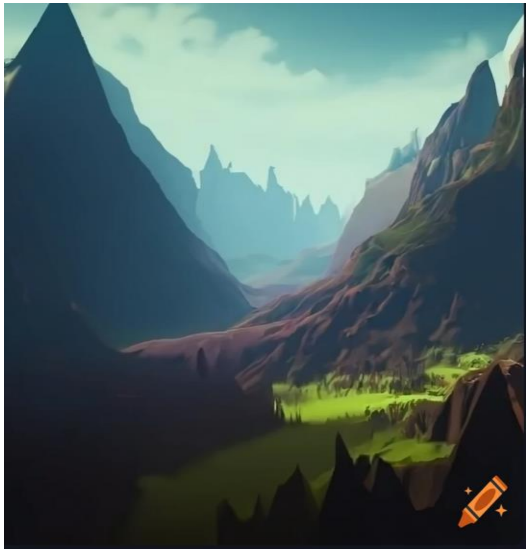
AI can generate realistic and unique textures and materials, which can be used to enhance the appearance of 3D models.

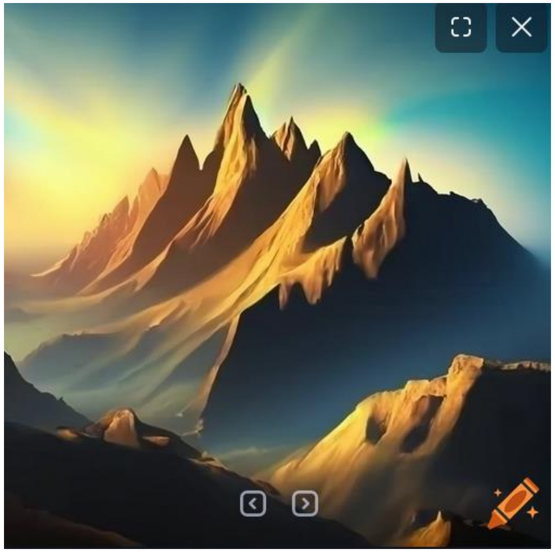


6- Create realistic landscapes

Procedural generation can be used to create vast and detailed landscapes with mountains, rivers, and forests







https://www.craiyon.com/image/20JZH78TQG6YpcB4wrowxA

7- Generate 3D cities

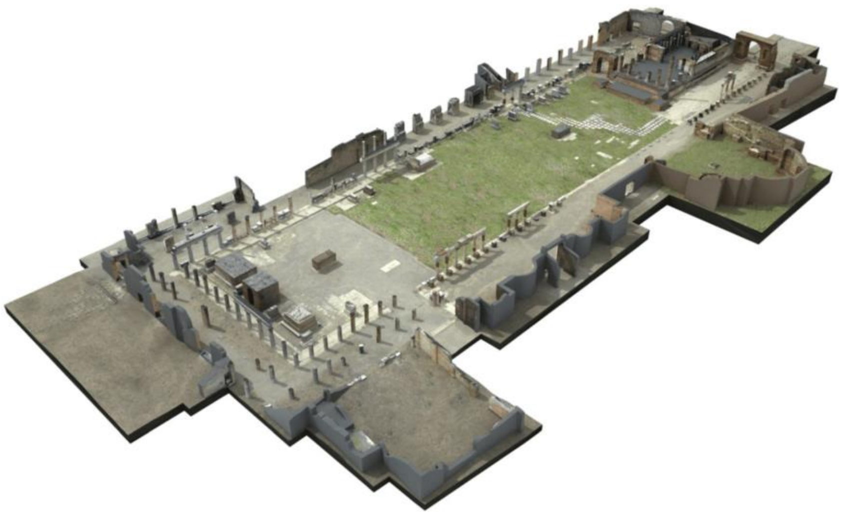
Procedural generation can be used to create realistic and diverse cities with buildings, streets, and cars.

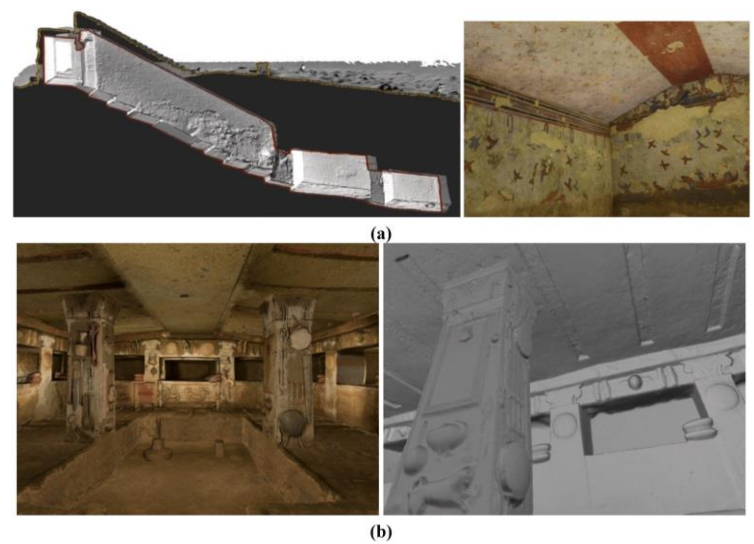
https://simteract.com/projects/city-generator/ , https://vizzio.ai/

8- Digital Heritage التراث الرقمي

Photogrammetry and Laser scanning in cultural heritage

Combine the accuracy of LiDAR with the detail and color of photogrammetry to create high resolution, high accuracy models of cultural heritage sites. The creation of accurate models and high resolution ortho-tiffs is essential in active conservation at cultural heritage sites. Detailed documentation of specific monuments aids in the active conservation and comparative analysis. Such analysis can be used to better understand how various factors (seismic activity, climate changes, wars, etc.,) have affected the monument or whole site.



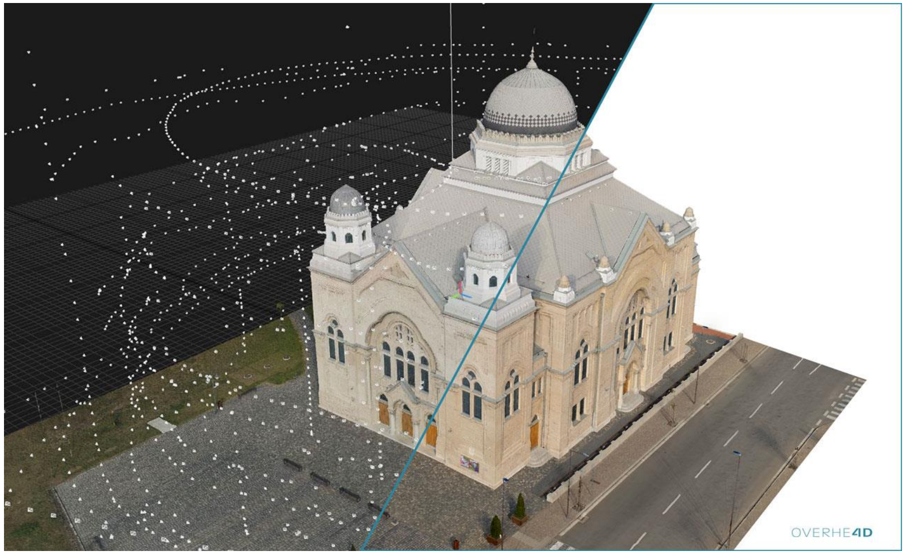


https://www.mdpi.com/2072-4292/3/6/1104 , https://www.capturingreality.com/cultural-heritage

للاطلاع
فقط

Example : **Creating Virtual Guide (Reality Capture)**

The exterior of the synagogue has been created from almost 4,000 photos, captured with the DJI Phantom 4 Pro drone and the Sony A7R II camera.



للاطلاع
فقط

Example : **Creating Virtual Guide**

Online View: https://sketchfab.com/3d-models/synagogue-lucenec-slovakia-lower-resolution-9826c734b27b440081ef1df7a284599d

Technical specification of the project

Exterior capture	3,500 drone images & 400 images ground photogrammetry
Interior capture	6,300 DSLR images, 300 drone images & 24 laser scans
Drone	DJI Phantom 4 Pro
Laser scanner	Faro Focus X330
Ground photogrammetry	Sony A7R II
Final model	7M tris & 11x 8k textures
Capture time	1 day for exterior, 1 day for interior

https://www.capturingreality.com/Virtual-Guide-with-RealityCapture

9- 3D Reconstruction

Point cloud

Point cloud can be constructed from RGB-D images. If you have a RGB-D scanning and also known the intrinsics of your scanning camera, you will be able to create the point clouds from the RGB-D image, by simply calculating the real world (x, y) with your camera intrinsics. It is called **camera calibration**. Hence, till now, you should know that the **RGB-D image are grid-aligned images while the point cloud is in a more sparse structure**.

3D Vision

Just like 2D problems, we would like to detect and recognize all the objects inside a 3D scan. Unlike the 2D images, what is the best inputting format of the data to make use of CNNs is the question.

https://medium.com/@jianshi_94445/an-introduction-towards-3d-computer-vision-71be8ce11956

10- Metaverse and Web3

The metaverse is a virtual world where people can interact with each other and with digital objects. Web3 is a new iteration of the internet that is built on blockchain technology. Research in the metaverse and Web3 is focused on:

- **Creating immersive and interactive virtual experiences:** This includes developing new ways for people to interact with each other and with digital objects in the metaverse.
- **Decentralizing the metaverse:** This means that the metaverse would not be controlled by any one company or organization.
- **Making the metaverse accessible to everyone:** Research is being done to make the metaverse affordable and easy to use for everyone.

thanks